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Precise GNSS-acoustic seafloor positioning with sound speed from global ocean analysis

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Abstract

The Global Navigation Satellite System–Acoustic (GNSS-A) combined positioning technique extends geodetic networks into the seafloor. Currently, GNSS-A can achieve static seafloor positioning accuracy at centimeter level. However, in practical operations, substantial time, manpower, financial and instrument resources are required to measure in situ Sound Speed Profiles (SSPs). This paper evaluates the feasibility of GNSS-A with alternative SSPs instead of in situ measurements. The GNSS-A positioning using three different SSPs are compared: the Munk empirical profile, the profiles from the HYbrid Coordinate Ocean Model (HYCOM) global ocean analysis product, and the in situ profiles. Compared with the in situ profile, the Munk SSP has little impact on the GNSS-A horizontal position (0.6 cm in root-mean-square, RMS) but introduces a large systematic error in the vertical position (10.3 cm in RMS), and the impact on the displacement velocity is at the mm/a level. When the HYCOM profile is substituted for in situ profiles, the impact on GNSS-A positioning is only 0.2 cm in the horizontal and 2.9 cm in the vertical, and the impact on displacement velocity is at the sub-mm/a level in the horizontal and mm/a level in the vertical. The HYCOM global ocean analysis SSPs can largely serve as a cost-effective substitute for in situ profiles in GNSS-A seafloor positioning, which is especially applicable to GNSS-A measurements using unmanned surface vehicles, for which full-depth SSP measurements are difficult. Therefore, when SSPs are selected, appropriate decisions should be made on the basis of specific GNSS-A application needs and conditions.

Keywords GNSS-acoustic, Seafloor geodetic positioning, Sound speed profile, Global ocean analysis, Munk profile, Error compensation

Introduction

A seafloor geodetic network provides important technical supports for terrestrial reference frames, ocean observations, and marine resource exploration. The seafloor transponder array is a powerful tool for monitoring the displacements and deformations in the boundaries and

interior of seafloor plates (Bürgmann & Chadwell, 2014; Sato et al., 2011), providing valuable insights into fault, seismic, volcanic, and tsunami studies (Gagnon et al., 2005; Fujimoto et al. 2014; Yokota et al., 2016). For the effective construction of a marine geodetic network, the precise positioning of marine geodetic points is essential.

Global Navigation Satellite System–Acoustic (GNSS-A) combined positioning technique, pioneered by Fred Spiess in the mid-1980s (Spiess, 1985; Spiess et al., 1998), is an essential tool in seafloor geodesy. With continuous improvements and developments over the past decades, GNSS-A has become a critical tool in the study of crustal deformations in deep ocean and nearshore areas (Bürgmann & Chadwell, 2014; Iinuma et al., 2021; Xu et al., 2005). The advances in kinematic GNSS positioning

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and acoustic ranging technologies since the mid-1980s have enabled accurate seafloor geodetic measurements. Long-term seafloor monitoring has revealed the plate movements in the Cascadia subduction zone, the Japan Trench, and the Peru–Chile Trench (Chadwell et al., 2002; Fujita et al., 2006; Gagnon et al., 2005), as well as intermittent ridge extensions at the Juan de Fuca Ridge (Chadwell et al., 1999).

GNSS-A positioning technique mainly includes two parts: the position and orientation of a sea surface vehicle, and the acoustic ranging from the surface vehicle transducer to the seafloor transponders. Currently, the kinematic position accuracy of the surface vehicle reaches approximately 2 cm horizontally and 3–6 cm vertically, and its heading accuracy reaches 0.01–0.001 degrees, and the pitch and roll accuracy reach 0.01 degrees (Geng et al., 2010; Watanabe et al., 2017). In comparison, the accuracy of underwater acoustic positioning still needs to be improved. Currently, the measurement error of underwater acoustic time delay is typically less than 10 microseconds, which translates to a ranging error of less than 1.5 cm (e.g., Han et al., 2016; Spiess et al., 1998). The lever-arm and angle of the transducer installation can be precisely measured using a total station or estimated and calibrated (Chen et al., 2019; Liu et al., 2006). The timing of the transducer can be provided by GNSS timing to nanosecond level, and the response delay of the seafloor transponder (including hardware delay) can be precisely calibrated to the millimeter level (Xu et al., 2005; Chadwell et al. 2008). These systematic errors can be effectively mitigated or eliminated. However, the in-situ sound speed measurement accuracy is at best approximately 1.5 cm/s to 5 cm/s (Bürgmann & Chadwell, 2014), and the in-situ measurement sampling cannot adequately represent the spatial–temporal changes of sound speed. Thus, besides the constraints of the observation geometry structure (Chen et al., 2020; Li et al., 2023b), the main factor affecting the positioning accuracy of GNSS-A is the sound speed error.

Oceanographic disturbances causing the variations in the underwater Sound Speed Structure (SSS) constitute one of the major error sources of GNSS-A. The GNSS-A campaign at one site is usually conducted with a research vessel in a survey period of several tens of hours. The net lateral bias in the GNSS-A position estimate can be mostly removed by conducting the observation for tens of hours to days. Currently, GNSS-A technique is capable of horizontal positioning with an accuracy at centimeter level with 24 h of data and likely subcentimeter-level precision after 3–4 d (Chadwell & Spiess, 2008; Fujita et al., 2006; Gagnon et al., 2005). Although the uncertainty of GNSS-A positioning data differs at each observation site, the standard deviation

in the horizontal components (σ) is empirically approximately 2.0 cm (1.5 cm at the best site) in the data of the GNSS-A Seafloor Geodetic Observation Array (SGO-A), which is provided by the Japan Coast Guard (JCG), and the vertical component is typically worse by a factor of 2–3 (Yokota et al., 2018). Efforts to construct seafloor geodetic networks have led to significant research on the key technologies involved, with GNSS-A achieving accuracies of approximately 5 cm (Chen et al., 2018; Yang et al., 2020). Further improvements were made by compensating for the systematic errors in the acoustic ranging measurements (Yang et al., 2021; Liu et al., 2023). To observe the temporal changes in crustal deformation, which has been actively studied in recent years (Yokota et al., 2021, 2022), a positioning accuracy of 1 cm or higher is needed.

The spatial–temporal SSS in seawater varies significantly due to the changes in temperature, salinity and depth, which has a substantial impact on acoustic measurements and positioning accuracy. Although global ocean observation programs, such as the Array for Real-time Geostrophic Oceanography (Argo) and the Transparent Ocean Program (TOP), have been implemented to collect marine data (Hayes et al., 1991; Stammer et al., 1991; Wu et al., 2020), the temporal resolution of these programs (e.g., Argo observations every ten days) does not meet the requirements for high-precision seafloor geodetic positioning (Roemmich et al., 2009). Therefore, the in situ measurements of Sound Speed Profiles (SSPs) have become the main tool in GNSS-A, either through direct measurements or from Conductivity–Temperature–Depth (CTD) profiler data. However, obtaining reference SSPs does not guarantee high-precision seafloor positioning, as variations in sound speed significantly affect acoustic positioning systems (Kido et al., 2008). For example, the impact of sound speed variation was studied early in GNSS-A and SSP observations conducted off Hawaii Island (Osada et al., 2003). Subsequently, time-variable sound speed inversion methods based on reference SSPs were developed to mitigate these effects (Fujita et al., 2006; Liu et al., 2023; Zhao et al., 2018). To describe sound speed variation more precisely, researchers have developed inversion methods based on cubic B-spline models (Ikuta et al., 2008), enhancing the SSS detail by incorporating the concepts of first- and second-order horizontal sound speed gradients (Yokota et al., 2019). Despite these advancements, the in situ measurement of sound speed fields is costly, labor intensive, and time consuming, limiting GNSS-A applications. For example, it can take several hours for one SSP measurement operation in the deep ocean of thousands of meters, and eXpendable CTD (XCTD) or eXpendable BathyThermograph (XBT) probes cannot be restored for reuse.

Therefore, there is a need to find a cost-effective alternative that can effectively replace in situ measurement of SSPs.

In addition to in situ measured SSPs, the primary method used to construct the current sound speed field is using a historical sound speed field in the region. The concept of ocean acoustic tomography can successfully invert the underwater sound speed field (Munk & Wunsch., 1979). This is followed by the introduction of an Empirical Orthogonal Function (EOF) analysis method for Sound Velocity Profiles (SVPs) (LeBlanc & Middleton, 1980), which has been widely adopted. Using horizontal coordinate constraints and the empirical relationship between the sound speed and water temperature, the three-dimensional coordinates of the seafloor transponder and the sound speed field can be obtained (Osada et al., 2003). The previous studies explored sound speed inversion methods independent of in situ measured Sound Speed Profiles (SSPs) for acoustic seafloor positioning. Yang et al. (2011) proposed a precise positioning method for underwater static positioning without profile measurement, in which the sound ray bending error is expressed as a quadratic function of time, and the sound speed correction and the coefficients of the quadratic function are estimated together with the coordinates of the static points. By constraining the difference between the theoretical and observed spectra of sound signals, a grid search algorithm can find the optimal solution for sound speed (Li et al., 2015). Recent methods used water depth and sound speed gradients for self-constrained positioning without in situ measured SSPs, though the positioning error reached 0.54 m, which was significantly less accurate than the methods using in situ profiles (Zhao et al., 2022). Recent research showed that the use of auto structured empirical SSPs can, to some extent, replace in situ measured SSPs. The impact on the standard deviation of horizontal coordinates was less than 3 mm, the impact on the standard deviation of vertical coordinates was 2–3 cm, and the effects on station velocity in the three dimensions were 0.0, 0.4, and -1.4 mm/a (Xue et al., 2023). The experimental positioning accuracy of this method meets operational requirements, but the calculation involves many parameters, and the optimization process is complex and time-consuming, which needs further verification for more GNSS-A stations. The strict real-time acoustic positioning method for geophones, which is based on sequential adjustment and Baarda outlier detection, can achieve a positioning accuracy of 20 cm even without in situ measured SSPs, which has significant practicality, especially in the field of seismic exploration (Ma et al., 2024). Recent study explored

the feasibility of employing the global gridded marine environment datasets (GGMED) as substitutes for in situ sound speed measurements, and for the highest accuracy GGMED, the positioning accuracy can reach 2.1 mm in the horizontal and 24.4 mm in the vertical (Li et al., 2024). These studies demonstrate the ability to achieve high-precision seafloor positioning without the in situ measured SSPs. Although there is a decrease in positioning accuracy, they highlighted the potential replacement of in situ measurements and reduction of the operational costs of seafloor positioning.

We propose precise GNSS-acoustic seafloor positioning with the sound speed constructed from global ocean analysis products. Early oceanographers employed mathematical interpolation and statistical methods to create objective analysis datasets. However, they were primarily generated by applying mathematical and statistical techniques to control the quality of observational data and interpolate them onto fixed temporal and spatial grids. Consequently, they exhibit certain limitations in accurately depicting the dynamical features and physical processes of the ocean. In contrast, data assimilation integrates ocean numerical models with observational data to generate global ocean analysis products. These products construct continuous spatiotemporal fields, thereby providing a more comprehensive and accurate representation of oceanic dynamics and processes (Cummings & Smedstad, 2013; Helber et al., 2013). The operational global ocean analysis and forecasting system leverages ocean numerical models as the dynamical framework and assimilates near-real-time spatiotemporally irregular observational data into the model through data assimilation techniques to optimally estimate the system state, enabling forecasts of ocean conditions across multiple spatiotemporal scales on a global scale. The operational global ocean analysis and forecasting system aims to accurately analyze and predict the evolving three-dimensional distribution of temperature, salinity, currents, and associated mesoscale features, such as fronts and eddies (Metzger et al., 2014). The operational global ocean analysis and forecasting system run in real-time can provide routine analyses and forecasts products (Barton et al., 2021). Our results show that GNSS-A with the SSPs from the HYbrid Coordinate Ocean Model (HYCOM) (Bleck 2002) global ocean analysis product are comparable to those with in situ profiles in terms of station position and velocity. These advancements are not only significant for promoting cost-effective seafloor geodesy but also crucial for realizing real-time seafloor geodetic applications. By reducing the costs of SSP measurements and improving the accuracy of models for sound speed variations, the performance of acoustic seafloor positioning technology can be enhanced, providing new opportunities for marine science research and related applications.

This paper adopts a GNSS-A positioning model based on Bayesian estimation theory, focusing on precise estimation of seafloor geodetic station coordinates and velocities with various constructed SSPs, including the Munk empirical profile, the constructed profile via global ocean analysis from the HYCOM product, and the in situ measured SSP. The main objective of this study is to analyze the positioning differences among different SSPs and explore how these differences affect the accuracy of underwater positioning. This research employs Bayesian methods to estimate parameters with different SSPs, aiming to increase the accuracy and reliability of seafloor positioning in various environmental conditions and observation time spans. By comparing and analyzing the impacts of different SSPs on positioning and velocity outcomes, this paper can elucidate the specific role of SSP variations in underwater positioning accuracy. It strives to utilize cost-effective and readily available SSPs that meet accuracy requirements, providing method and practical reference for improving low-cost seafloor positioning technology.

Methodology

GNSS-A underwater positioning systems typically consist of two parts: the sea surface vessel segment and the underwater station segment. Here, we adopt the GNSS-A positioning method and the GNSS-Acoustic Ranging combined POsitioning Solver (GARPOS) software (version 1.0.0) described in Watanabe et al., (2020, 2021). In the surface vessel segment, the survey vessel follows a planned survey line, using GNSS kinematic positioning and an inertial measurement unit to determine the vessel's position and orientation. The position of the shipborne transducer, which is crucial for accurate underwater measurements, is then determined by considering the lever arm vector between the GNSS antenna and the acoustic transducer mounted on the vessel.

In the underwater segment, acoustic ranging is utilized, involving the transducer's location, the propagation time, and the sound speed field to determine the geometric distance between the transponder and the transducer. Using these data with respect to the GNSS reference coordinate system facilitates the establishment of a unified datum for both surface and subsurface environments. In the context of underwater acoustic ranging, given the position of the j -th seafloor transponder, $\mathbf{X}_j$, and a four-dimensional sound speed field represented as $V(e, n, u, t)$, the underwater acoustic time delay can be formulated as (Chadwell & Sweeney, 2010):

$$T_i = T_{i,c}(\mathbf{P}(t_{i+}), \mathbf{P}(t_{i-}), \mathbf{X}_j, V(e, n, u, t)) \quad (1)$$

where t_{i+} and t_{i-} respectively denote the transmission and reception times of the i -th acoustic signal, whereas

$\mathbf{P}(t_{i+})$ and $\mathbf{P}(t_{i-})$ represent the positions of the transducer at these two epochs.

Owing to the influence of seawater temperature, salinity, and pressure on the speed of sound in the ocean (Wilson & Wayne, 1960; Del Grosso, 1974), it is assumed that the marine sound speed structure is horizontally stratified and varies over time. Additionally, oceanic factors such as currents, tides, and internal waves result in horizontal disturbances in the sound speed field (Liu et al., 2021). Therefore, the four-dimensional sound speed field can be divided into static SSPs and dynamic disturbances:

$$\begin{aligned} T_{i,c}(\mathbf{P}(t_{i+}), \mathbf{P}(t_{i-}), \mathbf{X}_j, V(e, n, u, t)) \\ = \exp(-\gamma_i) \cdot \tau_i(\mathbf{P}(t_{i+}), \mathbf{P}(t_{i-}), \mathbf{X}_j, V_0(u)) \end{aligned} \quad (2)$$

In this context, $V_0(u)$ represents the reference SSP. The variable τ_i denotes the acoustic signal propagation time, which is calculated through ray tracing on the basis of the reference SSP. The parameters γ_i and $\mathbf{X}_j$ are the variables to be estimated. The expression $\exp(-\gamma_i)$ represents the difference ratio of the actual propagation time to the reference time caused by spatiotemporal disturbances in the sound speed field. Since the sound speed field is continuous in both time and space, a continuous scalar field, Γ , can be introduced from which correction coefficients are derived. The variable γ_i is defined as $\gamma_i \equiv \frac{1}{2} \sum_{l=i+}^{i-} \Gamma(t_l, \mathbf{P}(t_l), \mathbf{X}_j)$, which represents the average value of the sound speed disturbances at the moments of signal transmission and reception. This function is referred to as the sound speed perturbation model (Ming et al., 2023):

$$\Gamma(t, \mathbf{P}, \mathbf{X}) \equiv \alpha_0(t) + \alpha_1 \cdot \frac{\mathbf{P}}{L_*} + \alpha_2(t) \cdot \frac{\mathbf{X}}{L_*} \quad (3)$$

In this description, L_* represents the characteristic depth of the transponder. The coefficients α_0 , α_1 , and α_2 are time-varying coefficients for each term, with α_0 accounting for changes in the SSP over time. Given that the vertical variations in $\mathbf{P}$ and $\mathbf{X}$ are much smaller than the horizontal ones, the vertical components of α_1 and α_2 can be neglected. Therefore, α_1 and α_2 can be simplified to two-dimensional vectors used to represent the changes in the horizontal gradients.

Let $T_{i,o}$ represent the time delay for the i -th measurement. By taking the logarithm of both sides of Eq. 2, a new observation equation can be derived. This transformation typically helps linearize the relationship between variables, making it easier for parameter estimation and statistical analysis (Watanabe et al., 2020):

$$\log(T_{i,o}/T_*) = \log(\tau_i(X_j|V_0)/T_*) - \gamma_i(\alpha_0, \alpha_1, \alpha_2|X_{j,0}) + e_i \quad (4)$$

In the formula, the left side of the equation represents the ratio of propagation delays, whereas the right side, from left to right, represents the ratio of acoustic signal propagation delays calculated with ray tracing, the difference ratio caused by sound speed perturbations, and observational error. Here, T_* denotes the characteristic propagation time. This structure systematically accounts for the various factors influencing the accuracy of acoustic measurements in underwater environments. The seafloor transponder position and the sound speed perturbations are solved for using the empirical Bayes approach.

The sound speed is dependent on temperature, salinity, and pressure, all of which vary with depth. As depth increases, these variables change, and the vertical variation of the speed of sound is much larger than the horizontal variation. Thus, $|\gamma_i| < 1$ should be valid when the reference SSP properly represents the field; in this case, the relationship between the estimated sound speed correction δV_i and the average sound speed $\bar{V}_0$ of the reference SSP is as follows:

$$\delta V_i = \gamma_i \bar{V}_0 \quad (5)$$

It should be noted that several observation models can effectively compensate for sound speed errors (Xue et al., 2024; Yokota et al., 2019). Specifically, a network solution observation model incorporating zenith acoustic delay parameters has been proposed, which decomposes the zenith acoustic delay into a temporal component and two horizontal components, thereby capturing both the temporal and spatial variations of the sound speed field (Xue et al., 2024). When estimated with the observations collected within appropriate time windows, this model has demonstrated centimeter-level positioning accuracy.

Owing to the spatial-temporal complex variations of the marine environment, SSPs have errors. To analyze the travel time delay errors caused by the sound speed errors, the observation equation can be simplified as a travel time delay equation that focuses on sound speed errors. This equation can be expressed as follows (Li et al., 2023a,):

$$T_{i,o} = T_i[X_j, V(u, \delta V)] + e_i \quad (6)$$

where X_j represents the position of the seafloor transponder, the function $V(u, \delta V)$ defines the sound speed at a specific depth u , and e_i represents the observational error.

$$V(u, \delta V) = V_0(u) + \delta V(u) \quad (7)$$

where $V(u, \delta V)$ is the SSP, $V_0(u)$ is the reference SSP, and $\delta V(u)$ is the sound speed disturbance.

Given that the sound speed in seawater exhibits a horizontally stratified structure and that the incidence angle between adjacent water layers is not zero, according to Snell's Law, the propagation time t for each layer can be expressed as follows:

$$t = \frac{\Delta u}{V(1 - (pV)^2)^{1/2}} \quad (8)$$

In this context, p represents Snell's constant, and the integral over each layer's time can provide the one-way time delay for sound propagation through the water column. The total one-way time delay for sound propagation can then be calculated as follows:

$$T_i = \int_{u_r}^{u_s} \frac{1}{V(u, \delta V) \cos[\theta(u)]} du \quad (9)$$

where u_s is the water depth at which the shipborne transducer sends signals, u_r is the water depth at which the seafloor transponder receives signals, and $\theta(u)$ is the angle of incidence at depth u . Since the errors caused by sound speed disturbances are relatively small, the Taylor series expansion method can be used for approximation (Li et al., 2023a):

$$\frac{1}{V(u, \delta V)} = \frac{1}{V_0(u)} - \frac{\delta V(u)}{V_0^2(u)} \quad (10)$$

Substituting into Eq. (9), we obtain:

$$\begin{aligned} T_i &= \int_{u_r}^{u_s} \frac{1}{V(u, \delta V) \cos[\theta(u)]} du \\ &= \int_{u_r}^{u_s} \frac{1}{V_0(u) \cos[\theta(u)]} du - \int_{u_r}^{u_s} \frac{\delta V(u)}{V_0^2(u) \cos[\theta(u)]} du \end{aligned} \quad (11)$$

where $\int_{u_r}^{u_s} \frac{\delta V(u)}{V_0^2(u) \cos[\theta(u)]} du$ is the travel time delay error caused by errors in the SSP.

Experiment data

To analyze the impact of SSPs on the accuracy of GNSS-A seafloor positioning, we selected four sites located in different geographical areas. As shown in Fig. 1, these sites are located in the Shikoku/Kii area (including the sites KUM2 and TOS2) and the Tohoku area (including the sites FUKU and MYGI). The sites KUM2 and TOS2 are located off the southern coast of Shikoku Island in southwestern Japan. This region lies near the epicenter of the 1946 Nankaido earthquake and along the Nankai Trough (Sagiya & Thatcher, 1999). The area is characterized by a strong Kuroshio current, which flows eastward regularly throughout this region. Owing to the

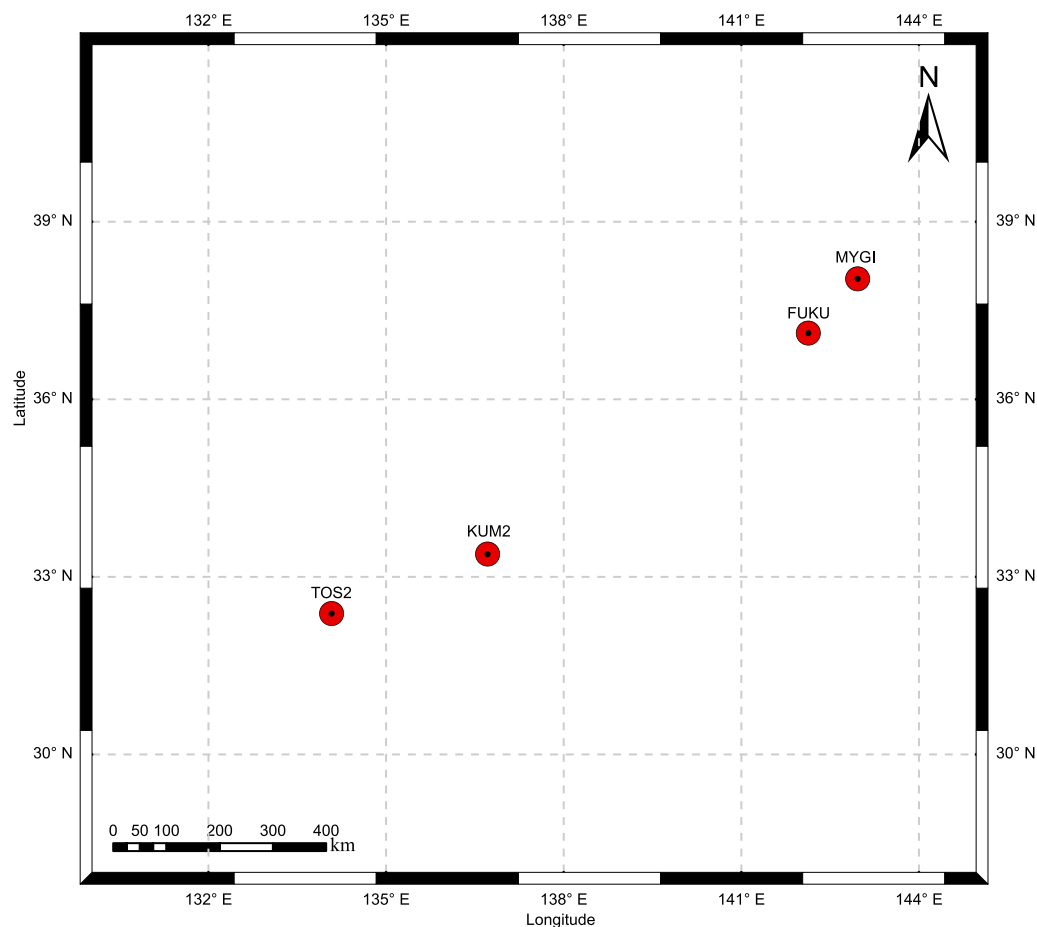


Fig. 1 Schematic diagram of GNSS-A site locations

perpendicular orientation of the Kuroshio current to this area's unidirectional oceanic structure, the oceanographic dynamics of the region are relatively uniform. In contrast, the sites FUKU and MYGI are located in northeastern Japan near the Japan Trench region, which experienced the 2011 Tohoku-Oki Earthquake, near the eastern coast of Miyagi Prefecture. Owing to the mixing of warm and cold currents, the oceanic structure of this area is more complex, featuring multiscale eddies and other phenomena (Kido et al., 2021; Wei et al., 2015). Specifically, the Kuroshio Current originates from the eastern Philippine Sea, travels northward along the eastern coast of Taiwan and the Ryukyu Islands, and then splits into two branches near the northeastern part of Honshu Island, Japan. One branch flows eastward into the Pacific Ocean, whereas the other branch moves northward to meet the cold, low-salinity Oyashio Current coming from the Bering Strait, forming the well-known Kuroshio–Oyashio confluence region. The Kuroshio Current is renowned for its high water temperature and salinity, whereas the Oyashio

Current is characterized by its low water temperature and salinity (Mensah et al., 2021; Yang et al., 2022). These two regions, influenced by different ocean currents, exhibit significant differences in their SSPs, providing an ideal case study to analyze and quantify the impact of SSPs on the accuracy of GNSS-A seafloor positioning in different marine environments. The GNSS-A and in situ sound velocity profile datasets used in this study are publicly available from the Japan Coast Guard (2022).

To analyze the accuracy of seafloor positioning, three types of SSPs derived with different methods were utilized: the in situ measured SSP, the profile using a global ocean analysis product, and the Munk empirical profile. First, for the in situ measured SSP, temperature and/or conductivity profiles were obtained from multiple measurements via CTD, XCTD, or XBT probes. The corresponding SSPs were subsequently calculated with an empirical formula based on the observed temperature and salinity profiles (Del Grosso, 1974; Watanabe et al., 2020).

Second, the Global Ocean Forecast System (GOFS) reanalysis (2012–2014) and operational analysis (2014–2020) products were obtained from the HYCOM open dataset (Cummings & Smedstad, 2013; Helber et al., 2013; Naval Research Laboratory, 2022), which is one of the most accurate global ocean analysis products. These GOFS products are generated by the United States Naval Research Laboratory (NRL) with the application of the HYCOM and the Navy Coupled Ocean Data Assimilation System (NCODA). The HYCOM is a widely used ocean numerical model that features a generalized vertical coordinate system. This system optimizes the distribution of vertical computational layers by configuring them as isopycnal in stratified regions, terrain-following in shallow coastal areas, and isobaric in the unstratified mixed layer. This advanced design of the vertical coordinate scheme enables HYCOM to effectively simulate a wide range of oceanic environments, particularly well-suited for modeling the domains that encompass both the open ocean and shallow or unstratified regions, with enhanced representation of the surface ocean and shallow coastal regions (Chassignet et al., 2007; Cummings et al., 2013). The GOFS reanalysis uses the advanced numerical model and data assimilation system to integrate the historical observational data from various types and sources. The GOFS operational analysis products are derived using all the observations that have become available since the last analysis. These encompass the surface observations from satellites, such as altimeter-derived Sea Surface Height (SSH) anomalies, Sea Surface Temperature (SST), and sea ice concentration, as well as in situ SST observations from ships and buoys. Additionally, they include temperature and salinity profile data from XBTs, CTDs, gliders, and Argo floats (Cummings & Smedstad, 2013; Metzger et al., 2014). We note that the typical daily availability of Argo profiles used in the data assimilation is only approximately 600 globally, which is relatively sparse. In our study area, the number of profiles generally exceeds the global average. We also note that SSPs from GNSS-A measurements in this study, which are available only several times a year, are significantly less frequent compared to other profiles. Consequently, these profiles do not contribute to the data assimilation (Cummings & Smedstad, 2013; Metzger et al., 2014). This system uses the 24-h model forecast as a first guess in a Three-Dimensional (3D) variational scheme and assimilates available observations, and generates a nowcast out through a seven-day forecast. The products are typically made available within two days following the model run. These products have a temporal resolution of 3 h and a spatial resolution of 0.08° in this study area. These products have 41 vertical levels (depth) of 0, 2, 4, 6, 8, 10, 12, 15, 20, 25,

30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, 200, 250, 300, 350, 400, 500, 600, 700, 800, 900, 1000, 1250, 1500, 2000, 2500, 3000, 4000, and 5000 m. The largest errors in these products are observed in the regions characterized by highly energetic and eddying western boundary currents, such as the Kuroshio (this study area), Gulf Stream, Agulhas Retroflexion, and Brazil-Malvinas Confluence, as well as along the Antarctic Circumpolar Current. However, the depth-averaged temperature error over the top 500 m remains below 0.5 °C across most of the global ocean (Metzger et al., 2014). On the basis of the temperature, salinity, and depth information recorded in this dataset, the SSPs for the respective areas were derived with an empirical formula (Chadwell et al., 2002; Del Grosso, 1974), which are referred to as the HYCOM SSPs hereinafter.

Finally, a Munk empirical SSP was constructed, characterized by a sound speed perturbation coefficient set to 0.007, with both the depth of the sound channel axis and the scale depth set to 1300 m, and a fixed sound speed of 1500 m/s at the sound channel axis.

$$c(z) = c_0[1 + \varepsilon(e^{-\eta} + \eta - 1)] = c_0 + c_0\varepsilon(e^{-\eta} + \eta - 1) \quad (12)$$

where $c(z)$ is the speed of sound at depth z , c_0 is the speed of sound at the sound channel axis, ε is the perturbation coefficient, $\eta = 2(z - z_1)/B$, where η represents the dimensionless distance relative to z_1 , z_1 is the depth of the sound channel axis, and B is the scale depth (Munk, 1974). Unlike the first two kinds of SSPs, this Munk model does not consider the specific geographical location of the sites, seasonal variations, or climatic factors. Sakic et al. (2018) proposed that smooth synthetic SSPs, called pseudo-profiles, can be obtained by least squares fitting of the coefficients ε and η of the Munk model to the measured profile, which successfully avoid numerical instabilities of the eikonal ray tracer with much more smoothed values than the raw SSPs, which have significant variations over depth ranges on the order of tens of meters. They also discovered that the in situ observed profiles can differ notably from their Munk pseudo-profiles, especially in the depth from 300 to 1000 m. By comparison, this study explores the impact of one uniform Munk SSP with fixed coefficients on seafloor positioning. In summary, the SSPs obtained from these three kinds of SSPs not only analyze the factors affecting seafloor positioning technology but also offer important references for the development of more efficient and cost-effective seafloor positioning technologies in the future.

In current practices of seafloor geodetic positioning, the commonly used method involves collecting acoustic travel time data from static seafloor stations during

cruises and measuring SSPs with probes such as SVP, CTD, XCTD, or XBT. Therefore, in this study, the positioning results obtained with this method are used as the standard reference values (i.e., "reference true") and are compared with the results obtained with the other two SSP methods to assess the impact of the SSPs on the positioning results. In the data downloaded from the HYCOM, owing to its resolution of 0.08° (which corresponds to a horizontal resolution of approximately 9 km), it is not possible to locate the specific site precisely. Consequently, we adopt the nearest-point strategy, selecting the HYCOM data point closest in distance to each site to represent the SSP of that site. The distances between the HYCOM data and the four sites are approximately 3.73 km, 2.91 km, 4.11 km, and 3.86 km, respectively. Despite the sound speed errors induced by this method, it remains a straightforward approach. If the HYCOM vertical levels deeper than the seafloor (constrained by the terrain) exhibit abnormal values, the values of the deepest two normal levels shallower than the seafloor are linearly fitted and extrapolated to the seafloor. If this simplified method can meet the positioning requirements, it can save a significant amount of computational resources in future applications such as real-time positioning.

Results and analysis

Sound speed profiles

Figure 2 shows the sound speed profiles at the four stations constructed from the in situ CTD profile, the HYCOM global ocean analysis product, and the Munk model. The black solid line represents the Munk model, the red solid lines represent the HYCOM global ocean analysis products, and the blue solid lines represent the in situ CTD profiles. As illustrated in Fig. 2, the three sound speed profiles have noticeable differences. The Munk profile, compared with the measured profiles, exhibits a large discrepancy, with sound speed differences of several tens of m/s, and it fundamentally does not consider any small-scale oceanic environmental effects, making its profile relatively simplistic. In contrast, the HYCOM profiles show smaller differences compared to the measured profiles, mostly fluctuating within 4–8 m/s, with the primary differences in the surface layer and the thermocline, where the maximum sound speed difference exceeds 10 m/s. Geographical location is a key factor contributing to the variation in sound speed. In particular, the sites FUKU and MYGI in the northeastern region are influenced by two different currents, making the temperature and salinity variations in these areas particularly sharp and complex. As shown in Fig. 2, the differences in sound speed at these two sites are significant, with discrepancies reaching 5.03 m/s and 2.93 m/s

in RMS, respectively. Conversely, the sites KUM2 and TOS2, which are influenced primarily by the Kuroshio Current, have relatively stable hydrological structures, with sound speed differences reduced to 2.31 m/s and 2.78 m/s in RMS, respectively. These variations can significantly impact the precise positioning of seafloor stations, which will be analyzed in the following section.

Displacement time series

Figure 3 displays the time series of displacements in the east, north and up components calculated with the in situ sound speed profiles. The displacements of each site are marked with circles to visualize the data points, and the linear trends are shown as solid lines, with the corresponding 95% confidence intervals represented by dashed lines, providing a clear assessment of the stability and precision of the positioning results. The displacement residuals show that when in situ SSPs are used, the GNSS-A seafloor positioning accuracy achieves centimeter-level both horizontally and vertically, which clearly reflects the characteristics of displacement changes in the three directions.

Figure 4 shows a comparison of the calculated displacement time series in the east, north, and up directions between the Munk and in situ sound speed profiles. Figure 4a shows that these two displacement trends at the four sites over approximately eight years are close to each other, and the RMSs of the slope differences are 1.5 mm/a in the horizontal direction and 3.7 mm/a in the vertical direction, with maximum differences of 2.5 mm/a in the horizontal direction and 4.3 mm/a in the vertical direction. Figure 4b shows that these two displacement trends at the four sites over approximately three years are close to each other, and the RMSs of the slope differences are 2.2 mm/a in the horizontal direction and 6.8 mm/a in the vertical direction, with maximum differences of 4.8 mm/a in the horizontal direction and 11.5 mm/a in the vertical direction. Figure 5 shows that these two GNSS-A positioning coordinates in the horizontal direction have differences of 0.6 cm in RMS and 2.9 cm in maximum, but those in the vertical direction have differences of 10.3 cm in RMS and 18.0 cm in maximum. The Munk SSP error has little effect on GNSS-A horizontal positioning but introduces an obviously large systematic error in vertical positioning.

The GNSS-A positioning with the Munk profile maintains consistency with that with the in situ profile in terms of displacement trends at the mm/a level and in terms of horizontal coordinates of 0.6 cm, but significant discrepancies of 10.3 cm in the vertical coordinates do not meet the requirements for centimeter-level precise positioning. Note that an eight-year displacement series

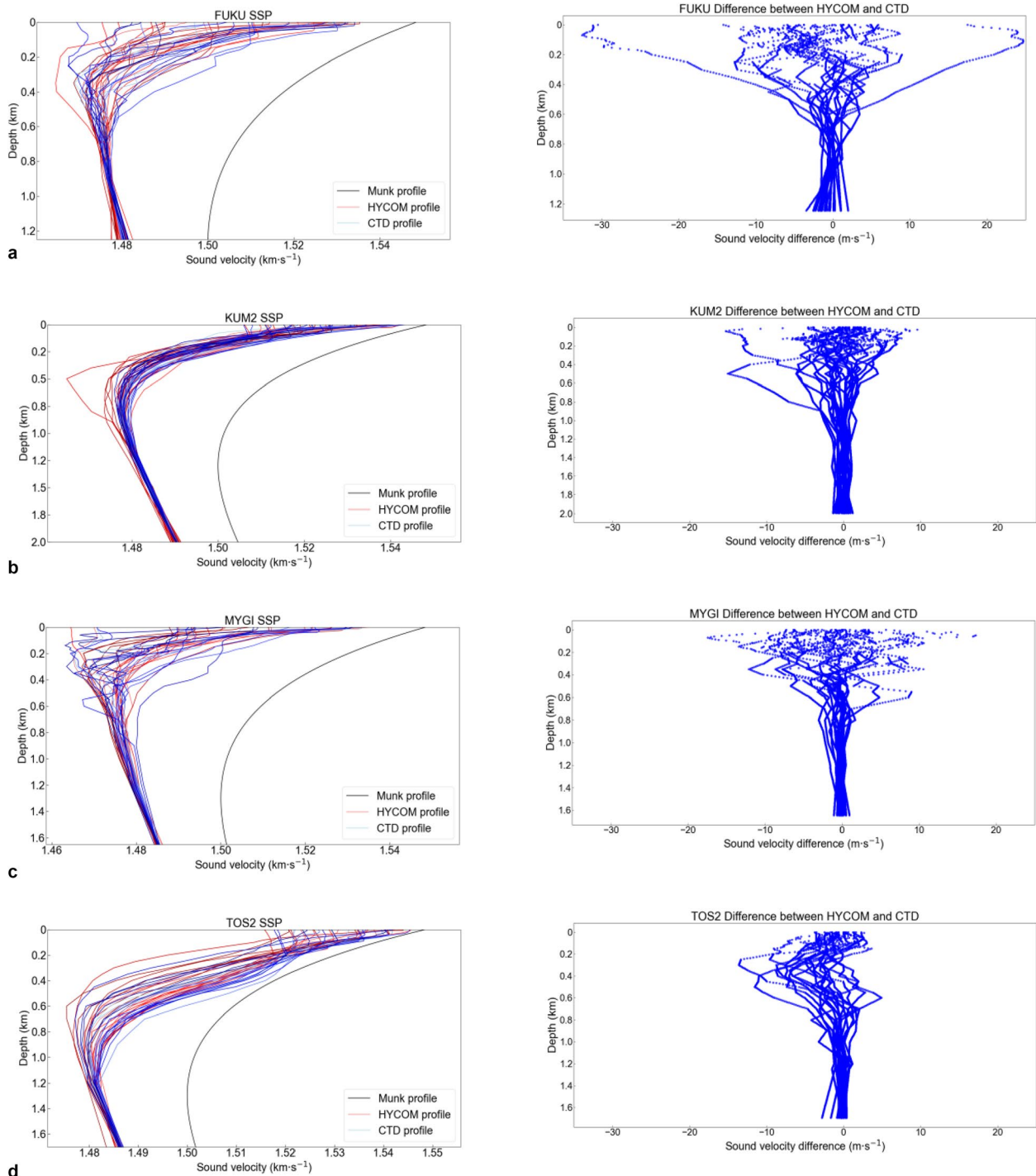
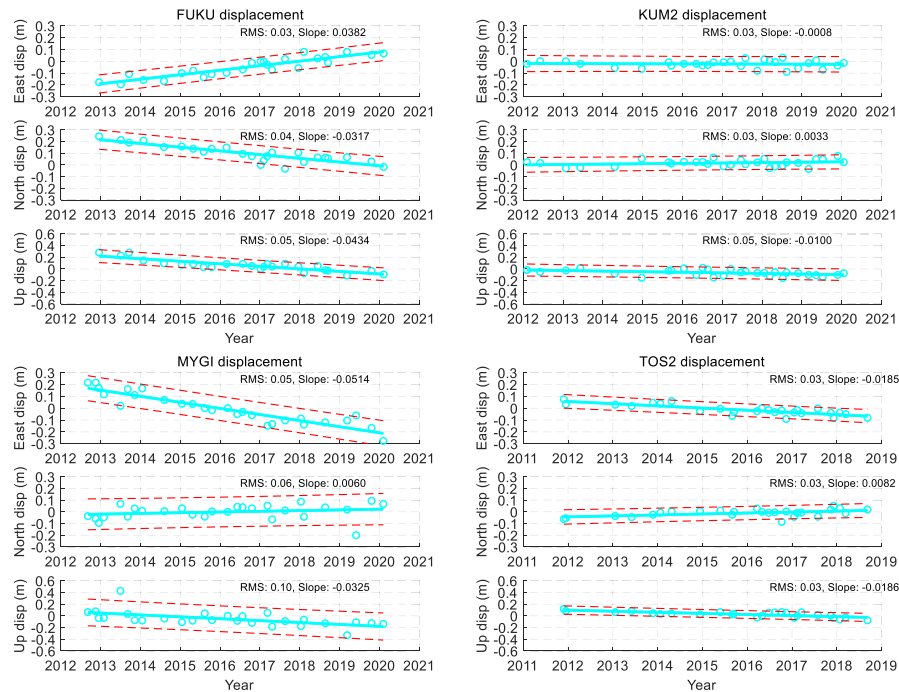


Fig. 2 Comparison of the in situ profile, HYCOM profile, and Munk profile sound speed at the four sites FUKU **a**, KUM2 **b**, MYGI **c**, and TOS2 **d**

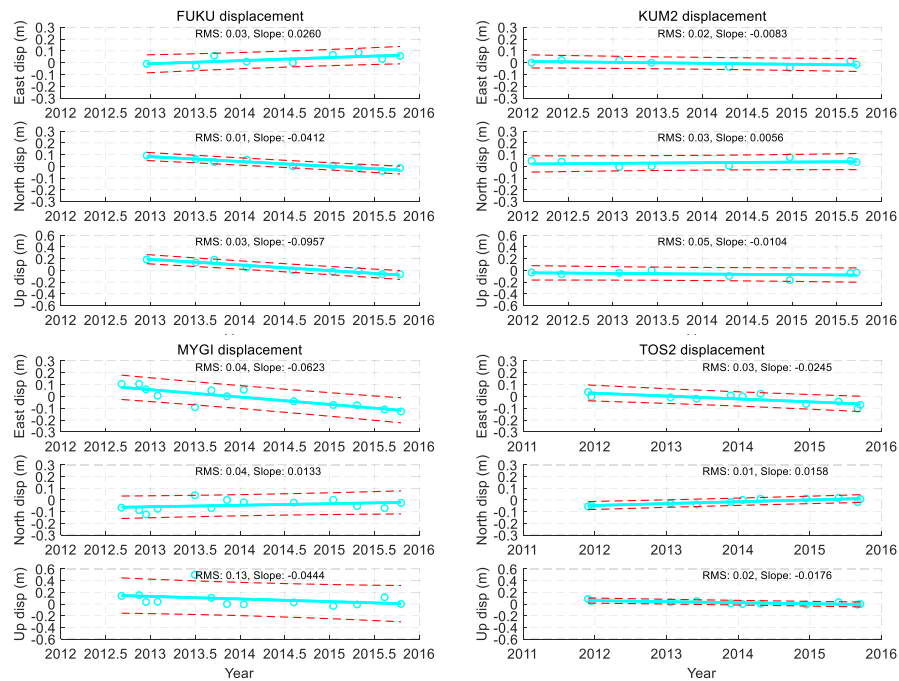
is long enough to mitigate the impact of large errors in individual positioning on the trend, and the three-year displacement series has greater trend differences.

Figures 6 and 7 show the GNSS-A positioning results using the HYCOM SSPs, as well as the differences from

those using the in situ SSPs. Figure 6a shows that these two displacement trends at the four sites over approximately eight years are close to each other, and the RMS of the slope differences is 0.3 mm/a in the horizontal and 2.7 mm/a in the vertical, with the maximum differences

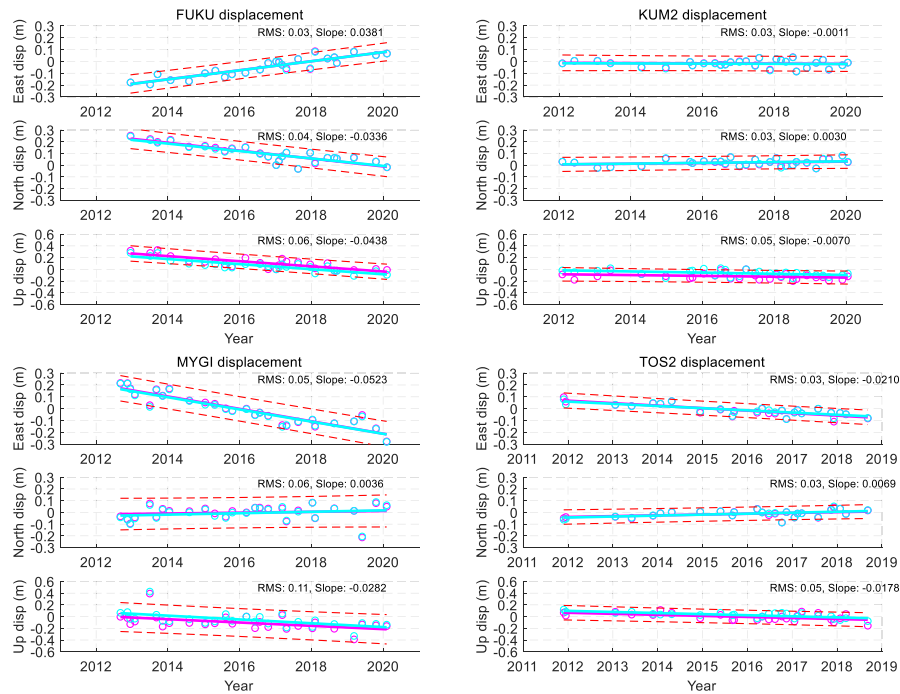


a Long-term displacements within situ sound speed profiles (from 2012–2020)

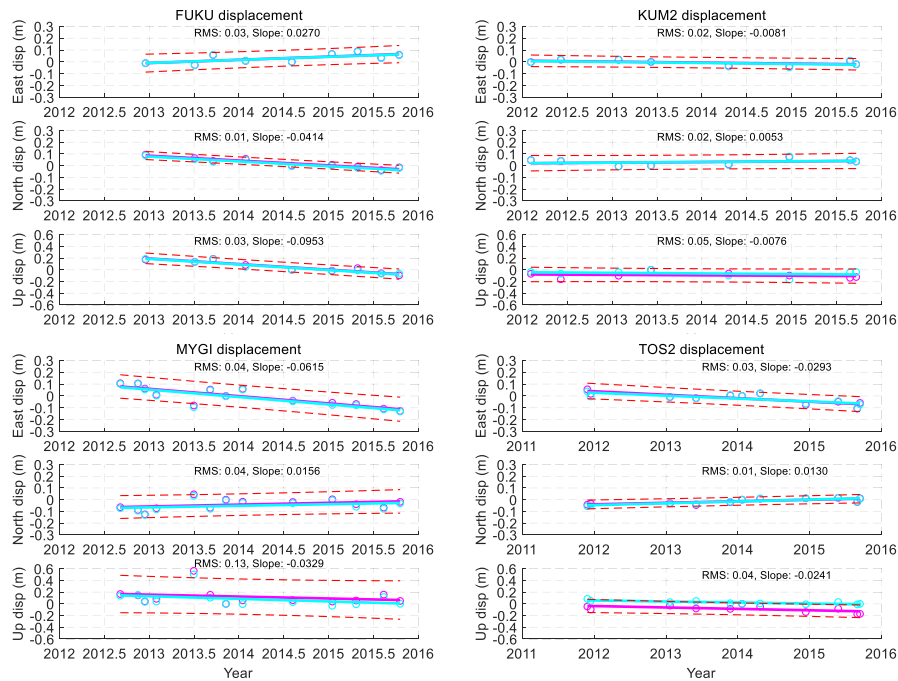


b Short-term displacements within situ sound speed profiles (from 2012–2015)

Fig. 3 Time series of displacements in the east, north and up components calculated with in situ sound speed profiles. The displacements of each site are marked with cyan circles to visualize the data points, and the linear trends are shown as cyan solid lines, with the corresponding 95% confidence intervals represented by red dashed lines. The RMS values of the displacement residuals with respect to the fitted linear trends (in meters) and the slopes of the fitted linear trends (in m/a) are shown in the upper panel



a Long-term displacements with the Munk empirical sound speed profiles (from 2012–2020)



b Short-term displacements with the Munk empirical sound speed profiles (from 2012–2015)

Fig. 4 Time series of displacements in the east, north and up components calculated with the Munk empirical sound speed profiles. The displacement of each site is marked with magenta circles to visualize the data points, and the linear trends are shown as solid magenta lines, with the corresponding 95% confidence intervals represented by red dashed lines. The RMS values of the displacement residuals with respect to the fitted linear trends (in meters) and the slopes of the fitted linear trends (in m/a) are shown in the upper panel. Note that the displacements and trends obtained with the in situ profiles are also depicted in cyan for comparison



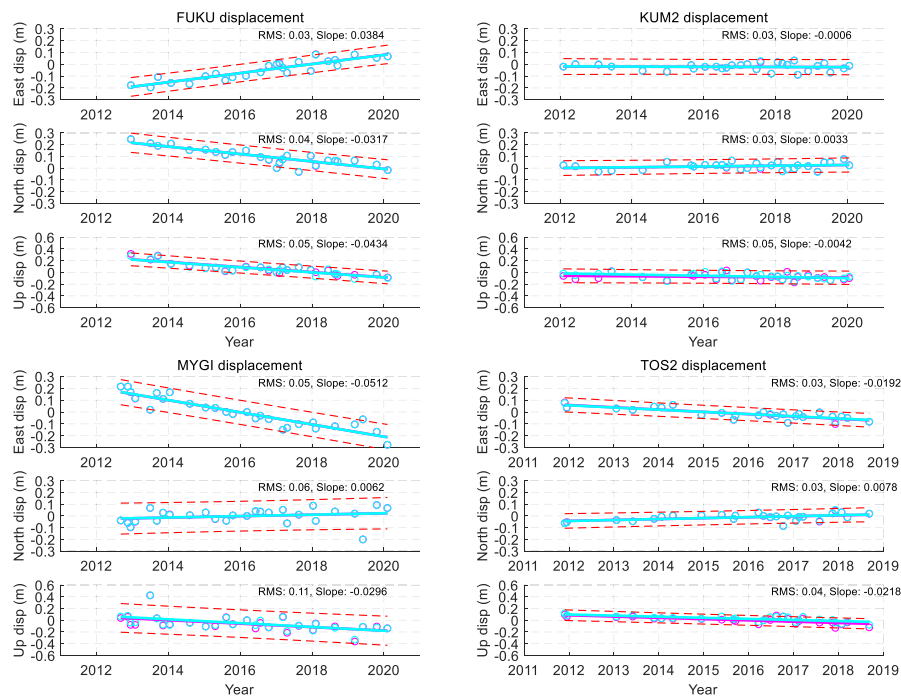
Fig. 5 Differences in displacements in the east, north and up components calculated using the in situ and Munk empirical sound speed profiles

of 0.7 mm/a in the horizontal and 3.2 mm/a in the vertical. Figure 6b shows that these two displacement trends at the four sites over approximately three years are close to each other, and the RMS of the slope differences is 0.6 mm/a in the horizontal and 10.2 mm/a in the vertical, with the maximum slope differences of 1.1 mm/a in the horizontal and 14.4 mm/a in the vertical. It can be observed that the linear fitting trends in all three directions at the four sites using the HYCOM SSPs are generally consistent with that using the in situ profile. Figure 7 shows that these two GNSS-A positioning coordinates in the horizontal have differences of 0.2 cm in RMS and of 1.8 cm in maximum, but those in the vertical have differences of 2.9 cm in RMS and of 9.1 cm in maximum.

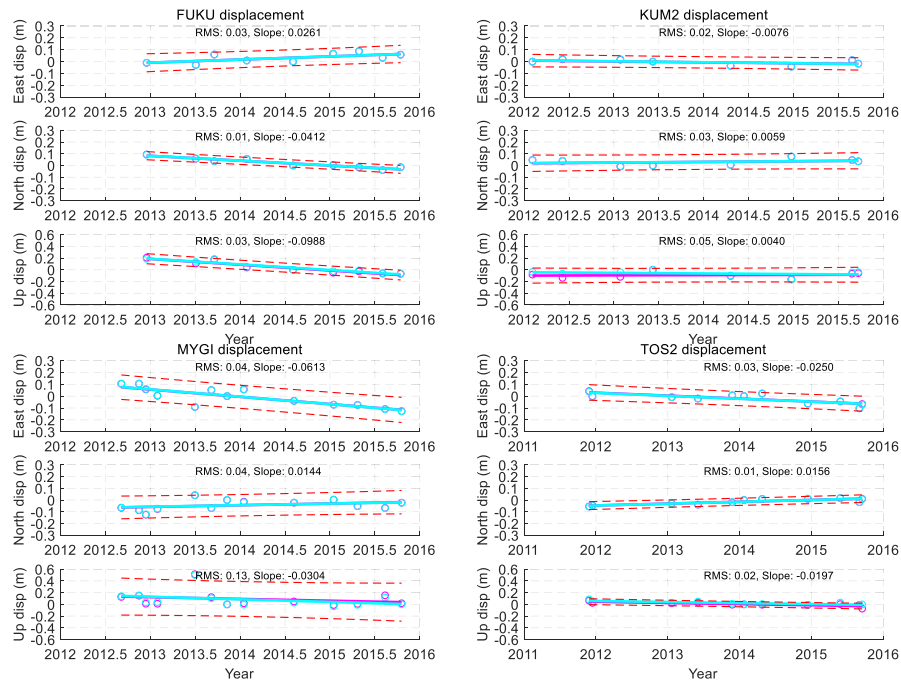
The GNSS-A positioning using the HYCOM profile maintains consistency with that using the in situ profile in terms of displacement trends at sub-mm/a level in the horizontal and mm/a level in the vertical, and in terms of horizontal coordinates of 0.2 cm, and the discrepancies of 2.9 cm in the vertical coordinates meet the current

GNSS-A centimeter-level precise positioning accuracy. This indicates that the HYCOM profiles can be adopted in practical GNSS-A operations, saving the enormous costs of in situ SSP measurements. Also note that an eight-year displacement series is long enough to mitigate the impact of large errors in individual positioning on the trend, and the three-year displacement series has greater trend differences. The GNSS-A positioning using the HYCOM SSPs show good consistency with that using the in situ profiles. The displacement trends are close to each other, and positioning discrepancies are at millimeter-level and centimeter-level in the horizontal and vertical direction, respectively.

Additionally, the displacement linear trend fitting residuals at sites TOS2 and KUM2 are smaller than those at FUKU and MYGI. This suggests that in the relatively stable marine environments, dominated by the Kuroshio current, the discrepancies between the SSPs from the global ocean analysis product and the in situ SSPs are smaller, and the TOS2 and KUM2 GNSS-A positioning



a Long-term displacements with the HYCOM SSPs (from 2012–2020)



b Short-term displacements with the HYCOM SSPs (from 2012–2015)

Fig. 6 Time series of displacements in the east, north and up components calculated using the HYCOM sound speed profiles. The displacement of each site is marked with magenta circles to visualize the data points, and the linear trends are shown as solid magenta lines, with the corresponding 95% confidence intervals represented by red dashed lines. The RMS values of the displacement residuals with respect to the fitted linear trends (in meters) and the slopes of the fitted linear trends (in m/a) are shown in the upper panel. Note that the displacements and trends obtained with the in situ profiles are also depicted in cyan for comparison

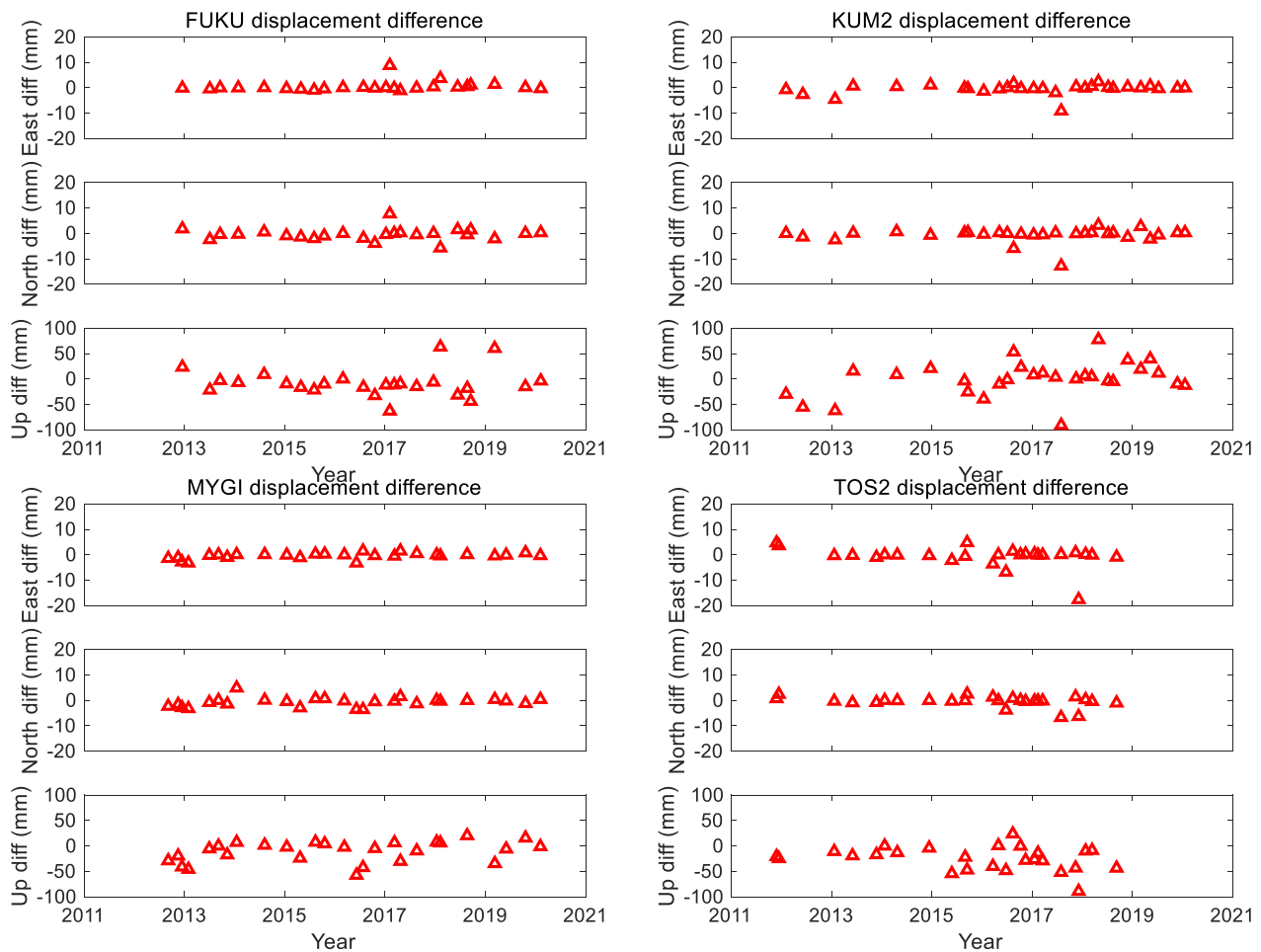


Fig. 7 Differences in displacements in the east, north and up components calculated with the in situ and HYCOM SSPs

errors are also smaller, thus resulting in smaller linear fitting residual. For the GNSS-A using the HYCOM SSPs, the average RMS value of the linear fitting residual is 2.3 cm in the horizontal components and 3.5 cm in the vertical component. In contrast, in complex marine environments such as in the Kuroshio and Oyashio currents meeting place, the discrepancies between the SSPs from global ocean analysis product and the in situ SSPs are greater, and the FUKU and MYGI GNSS-A positioning errors are larger, thus resulting in larger linear fitting residuals. For the GNSS-A using the HYCOM SSPs, the average RMS value of the linear fit residuals is 3.0 cm in the horizontal components and 8.0 cm in the vertical component.

Tables 1 and 2 show the GNSS-A-derived displacement velocities with three different SSPs in observation periods of approximately 8 years and 3 years, respectively. Generally, the shorter observation period of 3 years leads to greater differences in displacement velocities than those of 8 years. As shown in Fig. 5,

the difference between GNSS-A positioning with the Munk profile and the in situ profile is 0.6 cm in the horizontal direction and 10.3 cm in the vertical direction. Despite the large differences of 10.3 cm in vertical positioning between the Munk profile and the in situ profile, the differences in overall displacement velocities are only at the mm/a level. This can be due to the consistent systematic error and the average effect of random errors over a long-time span, which compensate for the positioning differences. As shown in Fig. 7, the difference between the GNSS-A position with the HYCOM profile and the in situ profile is 0.2 cm in the horizontal direction and 2.9 cm in the vertical direction, indicating that the position with the HYCOM profile is almost identical to that with the in situ profile considering the current centimeter-level GNSS-A positioning accuracy. The displacement velocity difference between the GNSS-A position with the HYCOM profile and the in situ profile is at the level of sub-mm/a in the horizontal direction and the

Table 1 Station displacement velocities in the East (E), North (N) and Up (U) directions from 2012 to 2020. (Unit: cm/a)

Site	Velocity (E) In situ	Velocity (E) Munk	Velocity (E) HYCOM
FUKU	3.82	3.81	3.84
MYGI	−5.14	−5.23	−5.12
KUM2	−0.08	−0.11	−0.06
TOS2	−1.85	−2.10	−1.92
Site	Velocity (N) In situ	Velocity (N) Munk	Velocity (N) HYCOM
FUKU	−3.17	−3.36	−3.17
MYGI	0.60	0.36	0.62
KUM2	0.33	0.30	0.33
TOS2	0.82	0.69	0.78
Site	Velocity (U) In situ	Velocity (U) Munk	Velocity (U) HYCOM
FUKU	−4.34	−4.38	−4.34
MYGI	−3.25	−2.82	−2.96
KUM2	−0.10	−0.70	−0.42
TOS2	−1.86	−1.78	−2.18

Table 2 Station displacement velocities in the east, north and up directions from 2012 to 2015. (Unit: cm/a)

Site	Velocity (E) In situ	Velocity (E) Munk	Velocity (E) HYCOM
FUKU	2.60	2.70	2.61
MYGI	−6.23	−6.15	−6.13
KUM2	−0.83	−0.81	−0.76
TOS2	−2.45	−2.93	−2.50
Site	Velocity (N) In situ	Velocity (N) Munk	Velocity (N) HYCOM
FUKU	−4.12	−4.14	−4.12
MYGI	1.33	1.56	1.44
KUM2	0.56	0.53	0.59
TOS2	1.58	1.30	1.56
Site	Velocity (U) In situ	Velocity (U) Munk	Velocity (U) HYCOM
FUKU	−9.57	−9.53	−9.88
MYGI	−4.44	−3.29	−3.04
KUM2	−1.04	−0.76	0.40
TOS2	−1.76	−2.41	−1.97

level of the mm/a in the vertical direction. Moreover, although the positioning coordinates are similar, the displacement velocity might be significantly impacted because of the lack of sufficient observational data to capture the velocity signal. When the HYCOM profile or in situ profile is used, the time span of the GNSS-A campaigns should be considered to avoid overinterpreting short-term observations.

Sound speed corrections

As shown in Fig. 8, using the Munk profile leads to a large estimated sound speed correction in GNSS-A processing,

reaching over 40 m/s. This is because the Munk profile ignored the regional marine environmental impacts. The time delay caused by these sound speed corrections, as shown in Fig. 9, range from 22 to 96 ms. The RMSs of the sound speed corrections are 30.43 m/s, 34.68 m/s, 17.57 m/s and 23.6 m/s for sites MYGI, FUKU, TOS2 and KUM2, respectively. The RMSs of the travel time delay correction are 58.9 ms, 48.8 ms, 36.1 ms and 55.2 ms for the MYGI, FUKU, TOS2 and KUM2 sites, respectively.

As shown in Fig. 10 and Fig. 11, the estimated sound speed corrections in GNSS-A processing with the HYCOM SSP are more than one order of magnitude

smaller than those generated with the Munk profile and are close to those with the in situ profile. The RMSs of the sound speed correction are 1.41 m/s, 3.24 m/s, 1.65 m/s and 1.15 m/s for the sites MYGI, FUKU, TOS2 and KUM2, respectively. Consequently, the time delay errors caused by sound speed errors are also over one order of magnitude smaller. The RMSs of the travel time delay correction are 4.7 ms, 2.8 ms, 3.5 ms and 2.7 ms for the sites MYGI, FUKU, TOS2 and KUM2, respectively. The dispersion of the sound speed corrections for the sites MYGI and FUKU is greater than that for the sites TOS2 and KUM2, which should be due to the greater modeling errors related to the complex marine dynamic environment. Geographically, the sites FUKU and MYGI are located in the Kuroshio–Oyashio confluence region. The mixing of these two currents in this area leads to a complex marine environment, including significant changes in temperature, salinity, and water currents. Conversely, the sites KUM2 and TOS2 are located in the areas

primarily influenced by the Kuroshio current, where the marine environment is relatively stable and uniform. The sound speed corrections at the sites FUKU and MYGI tend to be greater than those at the sites TOS2 and KUM2, which is due to the more intense marine environmental changes in the Kuroshio–Oyashio confluence region, leading to more heterogeneity and variability in the SSP, thus necessitating larger corrections for accurate sound speed estimation. However, there are exceptions, such as some sound speed corrections at the site TOS2 being higher than those at the sites FUKU and MYGI, which could be due to specific marine conditions during data collection.

These results highlight that the type of SSP selected has a significant effect on the extent of corrections needed when GNSS-A seafloor positioning is conducted. Owing to its coarser resolution and inaccuracies, the Munk SSP requires larger corrections to adapt to the actual marine environment. Its positioning results differ from

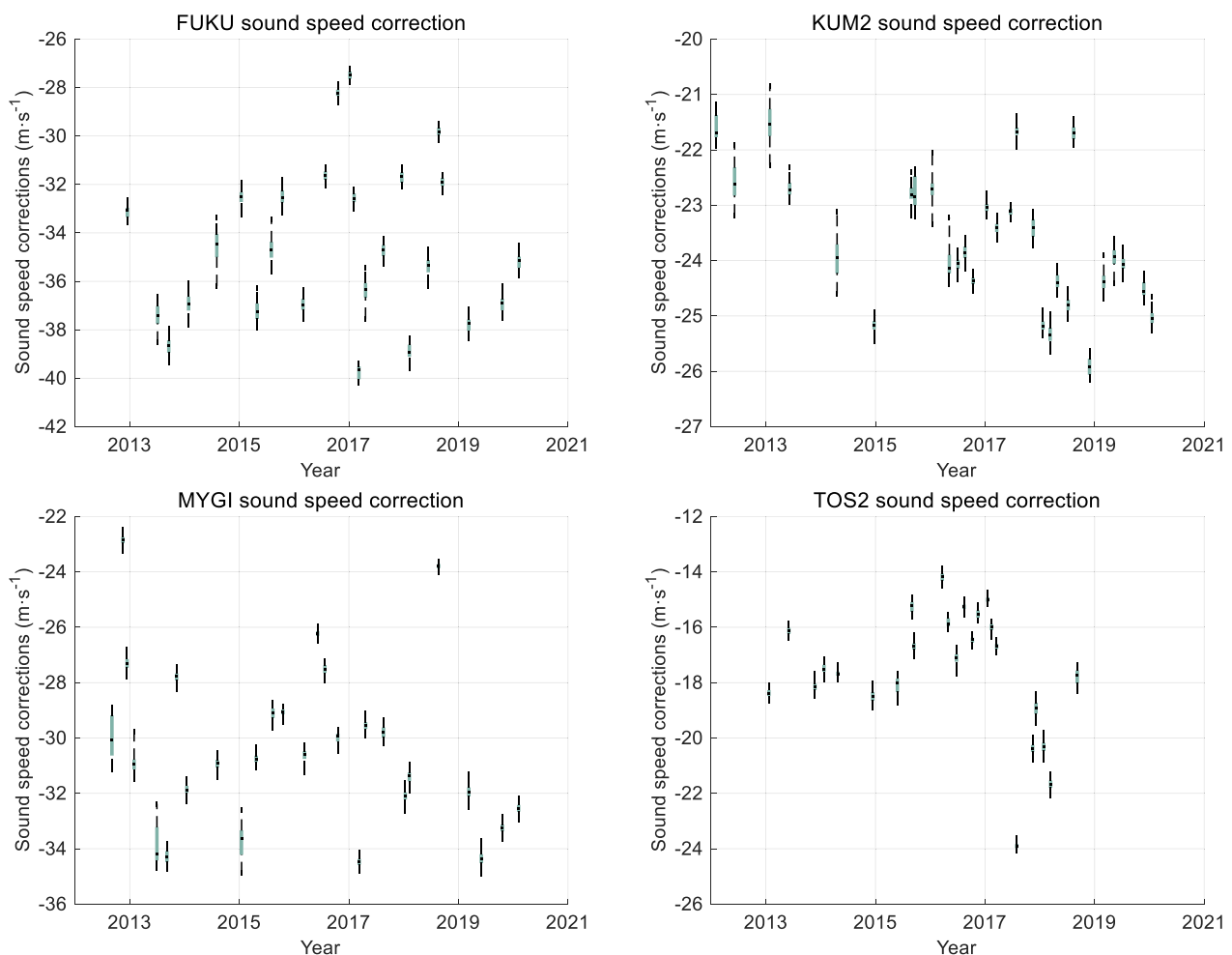


Fig. 8 Estimated sound speed corrections in GNSS-A processing with the Munk SSP. The boxes denote Interquartile Ranges (IQRs), the inside black lines denote their medians, and the outside black lines denote the maximum and minimum values

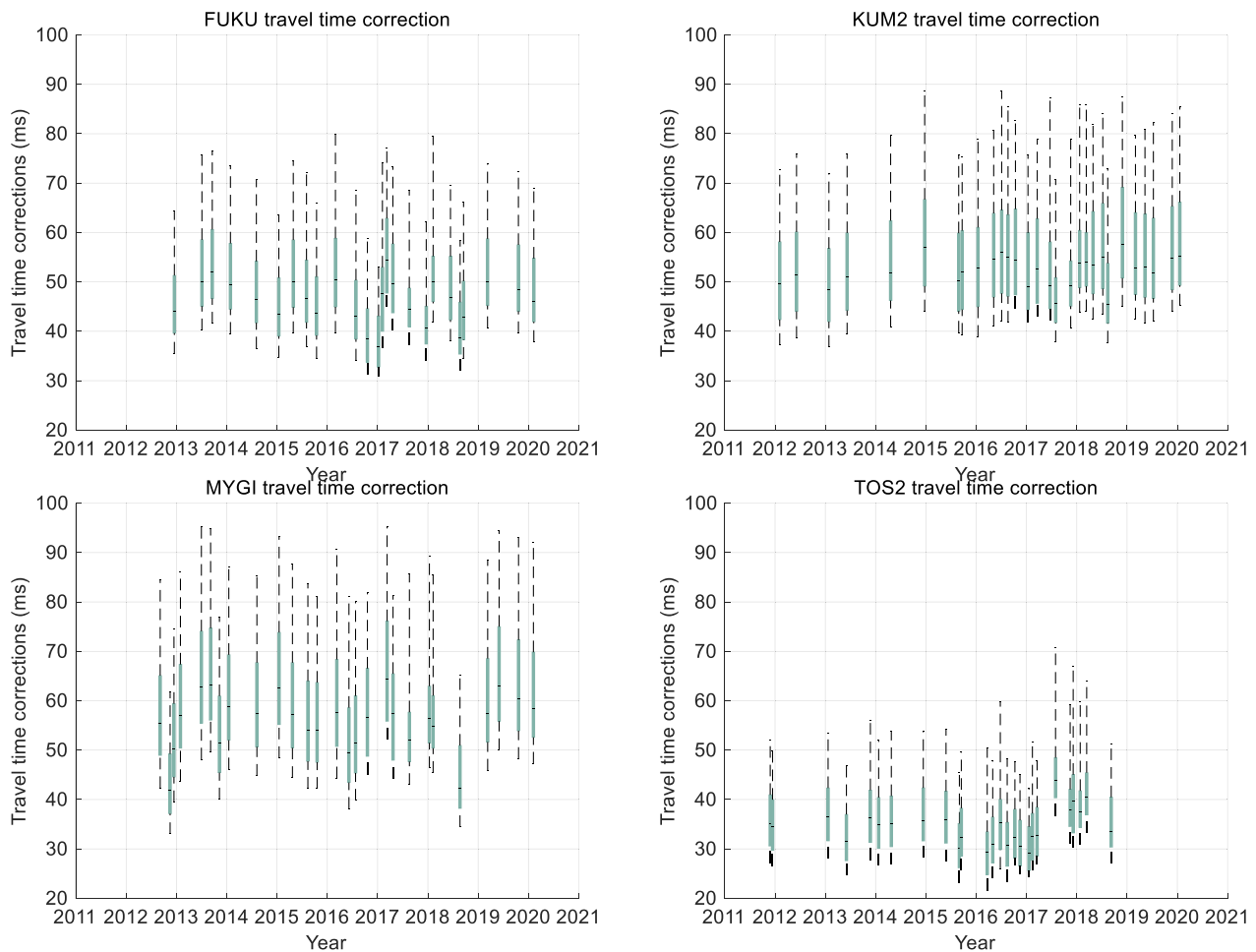


Fig. 9 Travel time corrections with the Munk SSP. The boxes denote interquartile ranges (IQRs), the inside black lines denote their medians, and the outside black lines denote the maximum and minimum values

the results with the in situ profile by 0.6 cm in the horizontal direction and 10.3 cm in the vertical direction. In contrast, the HYCOM profile, as a more accurate and adaptable representation, requires less corrections. Its positioning results differ from the results with the in situ profile by 0.2 cm in the horizontal direction and 2.9 cm in the vertical direction. A comparison of the sound speed error and time delay error caused by using the HYCOM and Munk profiles clearly reveals that the greater these errors are, the greater their impact on the positioning accuracy will be.

Time delay residuals

Despite the significant differences in the sound speed corrections and positioning results among the three SSPs used, they exhibit similar sound travel time delay residuals as shown in Fig. 12. This suggests that even if these SSPs vary in accuracy, they still perform comparably in GNSS-A in terms of adjusting the sound travel time in

the test conditions. This similarity in time delay residuals indicates that GNSS-A processing mitigates and compensates for the impact of sound speed inaccuracies on the adjusted sound travel time.

In summary, the HYCOM SSPs can essentially replace the in situ measured profiles for GNSS-A seafloor positioning. Although there are slight shortcomings in the vertical direction, the results in the horizontal direction are basically consistent with the in situ measured profiles, meeting the positioning requirements in practical operations. Moreover, since HYCOM data are easier to obtain, using HYCOM profiles to replace in situ measured profiles can significantly save manpower and resources and reduce costs.

Conclusions

GNSS-A positioning technology is an essential tool for seafloor geodesy. Currently, GNSS-A can achieve centimeter-level seafloor static positioning accuracy. However,

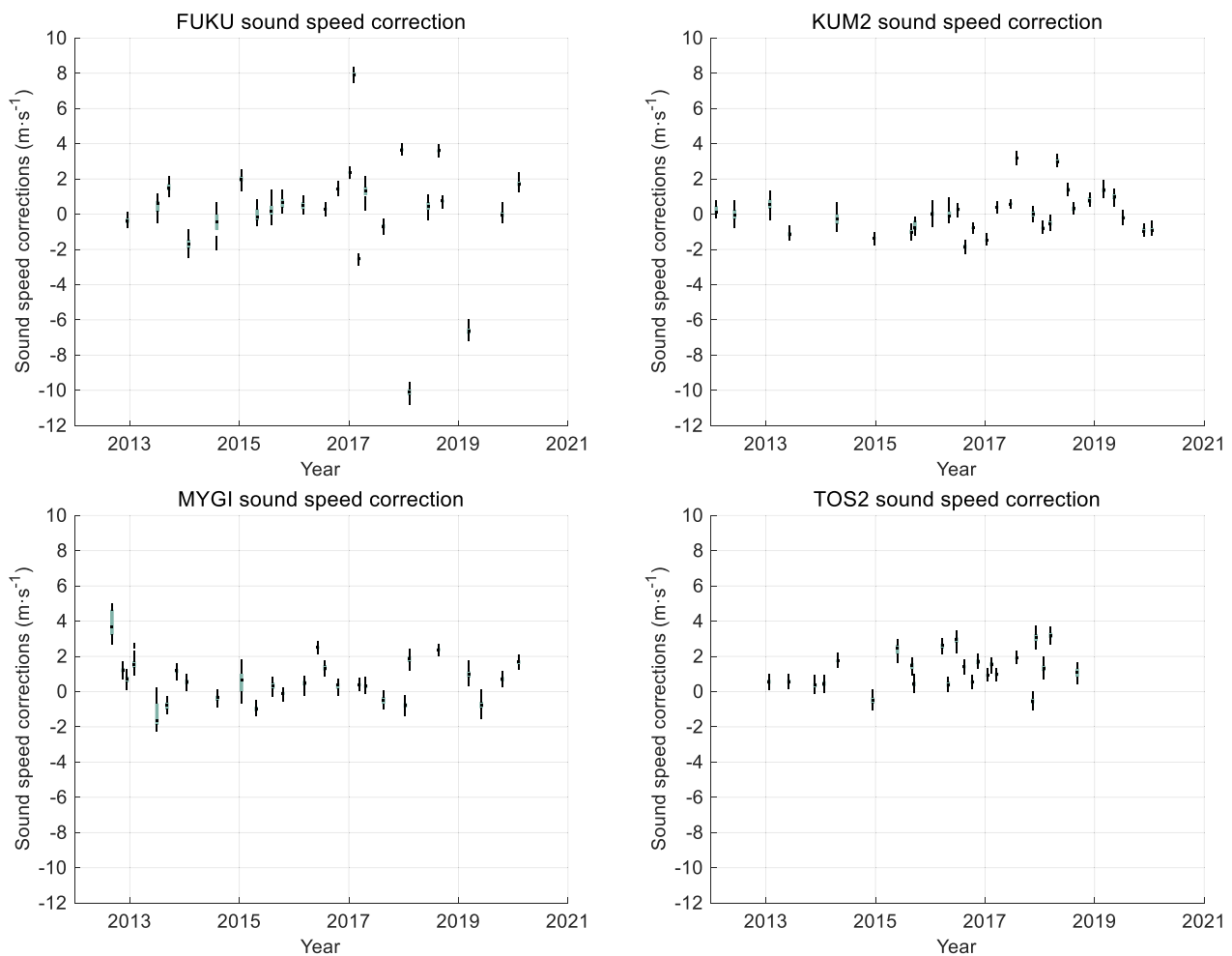


Fig. 10 Estimated sound speed corrections in GNSS-A processing with the HYCOM SSP. The boxes denote interquartile ranges (IQRs), the inside black lines denote their medians, and the outside black lines denote the maximum and minimum values

in practical operations, substantial time, manpower, financial and instrument resources are required to measure in situ SSPs. The GNSS-A campaigns usually adopt the designed geometrically symmetrical ship lines, and the GNSS-A seafloor positioning should compensate for some errors in the reference SSPs. This study evaluates the feasibility of GNSS-A seafloor positioning using the empirical Bayesian method with the Munk empirical profile and the profiles from the global ocean analysis product, instead of the in situ measured profiles. This study systematically analyzes the discrepancies in seafloor geodetic parameters (position/velocity), sound speed corrections, and travel time residuals among different SSPs used.

Compared with the in situ profile, the Munk SSP has little effect on the horizontal positioning of GNSS-A (0.6 cm in RMS) but introduced a large systematic error in the vertical positioning (10.3 cm in RMS). The GNSS-A positioning with the Munk profile remains consistent

with that with the in situ profile in terms of the displacement trend velocity at the mm/a level. While the Munk SSP can provide GNSS-A seafloor positioning, because of its inability to represent the actual regional environment, it significantly differs from in situ measured profiles, leading to vertical positioning accuracy that does not meet operational requirements. In applications requiring high precision, such as oceanographic research and seafloor exploration, the Munk SSP can provide reasonable displacement trend velocities, but it may not meet the vertical accuracy requirements.

Compared with the in situ profile, the HYCOM SSP achieves a GNSS-A horizontal position of 0.2 cm and a vertical position of 2.9 cm in the RMS and maintains consistency in terms of displacement trend velocities at sub-mm/a levels in the horizontal direction and mm/a levels in the vertical direction, meeting the current GNSS-A positioning accuracy at centimeter level. We can conclude that HYCOM SSPs can largely serve as an

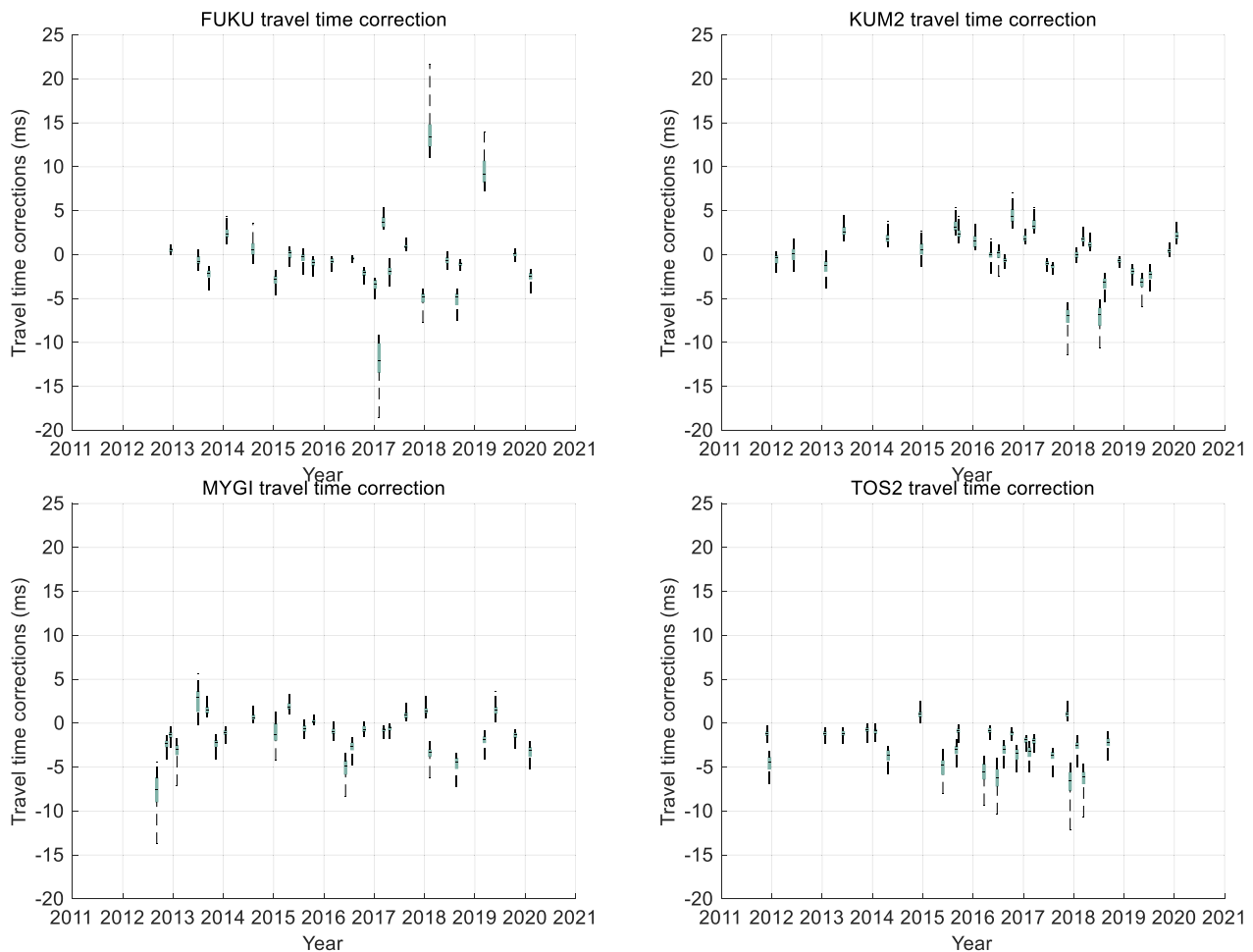


Fig. 11 Travel time delay corrections with the HYCOM SSP. The boxes denote IQRs, the inside black lines denote their medians, and the outside black lines denote the maximum and minimum values

effective substitute for the in situ measured profiles in GNSS-A seafloor positioning, meeting the positioning requirements for most practical operations. Considering the convenience and cost-effectiveness of obtaining HYCOM data, using HYCOM profiles instead of in situ measured profiles can significantly save manpower, instrument and ship time resources, reducing the costs associated with GNSS-A surveying and seafloor positioning, which is especially applicable to GNSS-A measurements using unmanned surface vehicles, for which full-depth SSP measurements are difficult. Therefore, when SSPs are selected, appropriate decisions should be made on the basis of specific application needs and environmental conditions.

This study provides some insight into the impact of SSPs on GNSS-A seafloor positioning, but there are still some issues that need to be addressed in future research. First, GNSS-A positioning with HYCOM SSPs shows a centimeter-level discrepancy in the vertical direction

compared with that with in situ measured profiles, which suggests that HYCOM profiles may not fully capture minor variations in the vertical direction. Notably, GNSS-A positioning with an in situ profile also has limitations for the applications requiring highly accurate vertical positioning. Second, a longer time span of GNSS-A data over eight years should provide more comprehensive and stable analytical results, helping accurately assess the long-term variations in SSPs and their impact on positioning accuracy. Theoretically, longer observation time could mitigate some errors, resulting in the determination of more precise trends in station displacement velocity. Third, for GNSS-A campaigns that last for days with geometrically designed symmetrical ship lines, errors should be greatly reduced, whereas for short campaigns that last for hours, these errors may be more pronounced, leading to the differences in positioning results. This means that in short campaigns, more accurate SSPs

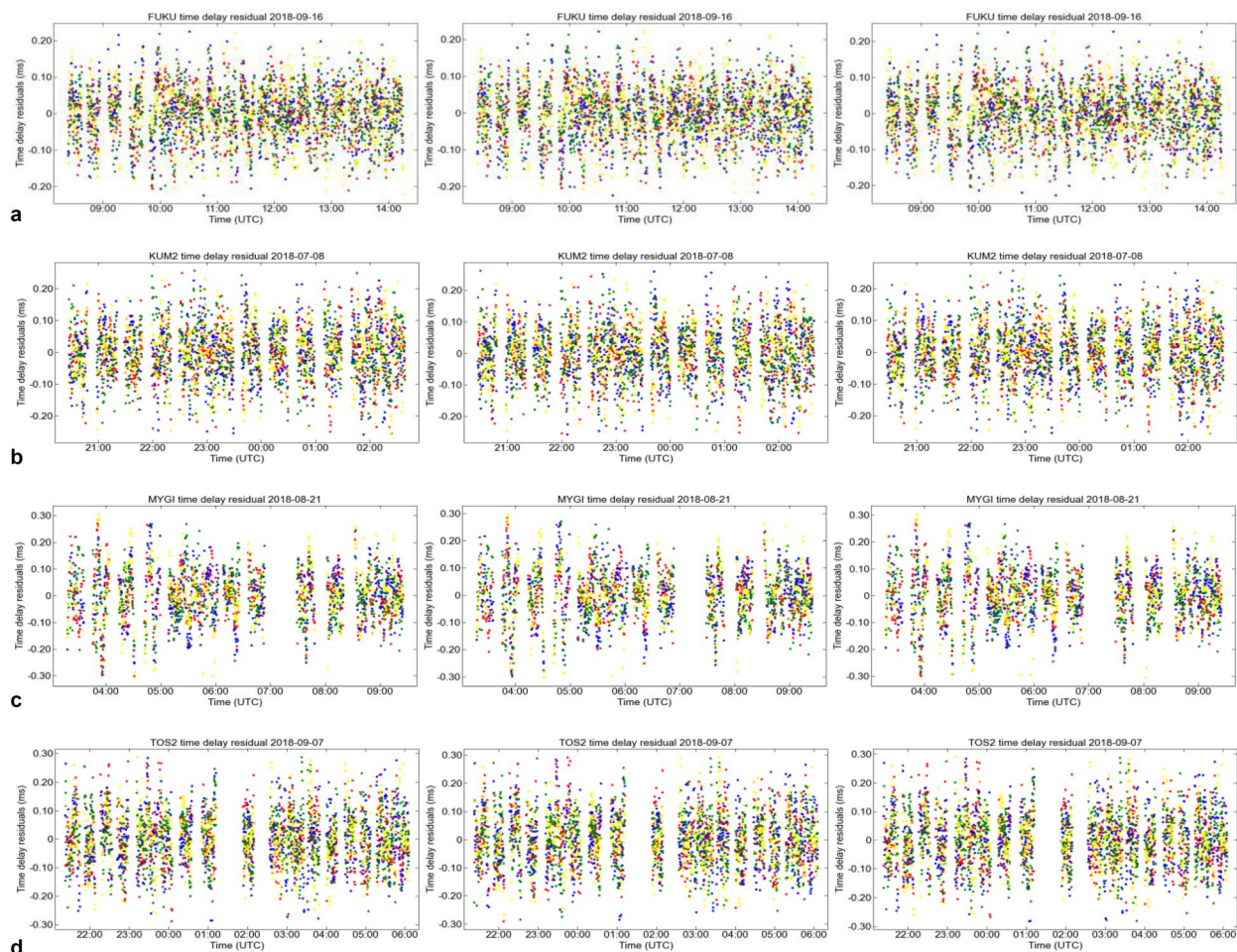


Fig. 12 Time delay residuals of one GNSS-A survey for the sites FUKU (a), KUM2 (b), MYGI (c), and TOS2 (d) with the in situ (left column), Munk (middle column), and HYCOM (right column) SSPs, respectively. The time delay residuals (in milliseconds) for individual transponders are color-coded for visual distinction

or fields, as well as advanced data processing techniques, are needed to ensure positioning accuracy.

Overall, although the SSP derived from global ocean analysis products can result in slightly poorer GNSS-A vertical positioning accuracy and there is room for improvement in certain aspects, using the global ocean analysis product for GNSS-A seafloor positioning is an effective method. Utilizing cost-effective and more readily available SSPs that meet accuracy requirements provides reference for low-cost seafloor geodesy. The accuracy of HYCOM itself is affected by various factors such as observations and models. The accuracy of HYCOM global ocean analysis product is uneven globally, and it is beneficial to further study other challenging regions of the highly energetic and eddying currents, as well as nearshore and polar regions. In addition, the SSP derived from other operational global ocean analysis and forecasting systems run in real-time can be used

to further study the performance of routine analyses and forecasts products in GNSS-A positioning. Future research can focus on improving positioning accuracy in the vertical direction, extending the observation time span, and exploring optimization strategies for positioning accuracy in different campaign time spans and operation conditions, especially for unmanned surface vehicles.

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Author contributions

YL and YL proposed the idea and method; YL and TS wrote the draft manuscript; TS conducted the experiment data analysis; GC and ML provide data analysis support; all the authors read and approved the final manuscript.

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Availability of data and materials

The GNSS-A and in situ SSP datasets were obtained from the website <https://zenodo.org/records/6417480> (Japan Coast Guard, 2022). The Global Ocean Forecast System (GOFs) reanalysis and operational analysis datasets were obtained from the HYbrid Coordinate Ocean Model (HYCOM) website <https://www.hycom.org/dataserver/gofs-3pt1/analysis> (Naval Research Laboratory, 2022). The GARPOS software is available at <https://zenodo.org/records/4522027> (Watanabe et al., 2021).

Declarations

Competing interests

The authors declare that they have no competing interests.

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References

- Barton, N., Metzger, E. J., Reynolds, C. A., Ruston, B., Rowley, C., Smedstad, O. M., et al. (2021). The Navy's earth system prediction capability: a new global coupled atmosphere-ocean-sea ice prediction system designed for daily to subseasonal forecasting. *Earth and Space Science*, 8, e2020EA001199. <https://doi.org/10.1029/2020EA001199>
- Bleck, R. (2002). An oceanic general circulation model framed in hybrid isopycnal-Cartesian coordinates. *Ocean Modelling*, 4(1), 55–88. [https://doi.org/10.1016/S1463-5003\(01\)00012-9](https://doi.org/10.1016/S1463-5003(01)00012-9)
- Bürgmann, R., & Chadwell, D. (2014). Seafloor geodesy. *Annual Review of Earth and Planetary Sciences*, 42(1), 509–534. <https://doi.org/10.1146/annurev-earth-060313-054953>
- Chadwell, C. D., Spiess, F. N., Hildebrand, J. A., & Dragert, H. (2002). Seafloor geodetic evidence of episodic spreading 25 km east of the Juan de Fuca Ridge. In *AGU Fall Meeting Abstracts* (Vol. 2002, pp. T22A-1130).
- Chadwell, C. D., Hildebrand, J. A., Spiess, F. N., Morton, J. L., Normark, W. R., & Reiss, C. A. (1999). No spreading across the southern Juan de Fuca Ridge axial cleft during 1994–1996. *Geophysical Research Letters*, 26(16), 2525–2528. <https://doi.org/10.1029/1999GL900570>
- Chadwell, C. D., & Spiess, F. N. (2008). Plate motion at the ridge-transform boundary of the south Cleft segment of the Juan de Fuca Ridge from GPS-Acoustic data. *Journal of Geophysical Research: Solid Earth*. <https://doi.org/10.1029/2007JB004936>
- Chadwell, C. D., & Sweeney, A. D. (2010). Acoustic ray-trace equations for seafloor geodesy. *Marine Geodesy*, 33(2–3), 164–186.
- Chassignet, E. P., Hurlburt, H. E., Smedstad, O. M., Halliwell, G., Hogan, P. J., Wallcraft, A. J., Baraille, R., & Bleck, R. (2007). The HYCOM (HYbrid Coordinate Ocean Model) data assimilative system. *Journal of Marine Systems*, 65, 60–83. <https://doi.org/10.1016/j.jmarsys.2005.09.016>
- Chen, H.-Y., Ikuta, R., Lin, C.-H., Hsu, Y.-J., Kohmi, T., Wang, C.-C., ... Ando, M. (2018). Back-arc opening in the western end of the Okinawa Trough revealed from GNSS/Acoustic Measurements. *Geophysical Research Letters*, 45, 137–145. <https://doi.org/10.1002/2017GL075724>
- Chen, G., Liu, Y., Liu, Y., & Liu, J. (2020). Improving GNSS-acoustic positioning by optimizing the ship's track lines and observation combinations. *Journal of Geodesy*, 94, 61. <https://doi.org/10.1007/s00190-020-01389-1>
- Chen, G., Liu, Y., Liu, Y., Tian, Z., & Li, M. (2019). Adjustment of transceiver lever arm offset and sound speed bias for gnss-acoustic positioning. *Remote Sensing*, 11(13), 1606. <https://doi.org/10.3390/rs11131606>
- Cummings, J. A., spsampsps Smedstad, O. M. (2013). Variational data assimilation for the global ocean. In *Data assimilation for atmospheric, oceanic and hydrologic applications (Vol. II)* (pp. 303–343). Berlin, Heidelberg: Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-35088-7_13
- Del Grosso, V. A. (1974). New equation for the speed of sound in natural waters (with comparisons to other equations). *The Journal of the Acoustical Society of America*, 56(4), 1084–1091. <https://doi.org/10.1121/1.1903388>
- Fujimoto, H. (2014). Seafloor geodetic approaches to subduction thrust earthquakes. *Monographs on Environment, Earth and Planets*, 2(2), 23–63. <https://doi.org/10.5047/meep.2014.00202.0023>
- Fujita, M., Ishikawa, T., Mochizuki, M., Sato, M., Toyama, S.-I., Katayama, M., Kawai, K., et al. (2006). GPS/Acoustic seafloor geodetic observation: method of data analysis and its application. *Earth, Planets and Space*, 58, 265–275.
- Gagnon, K., Chadwell, C. D., & Norabuena, E. (2005). Measuring the onset of locking in the peru-chile trench with GPS and acoustic measurements. *Nature*, 434(7030), 205–208. <https://doi.org/10.1038/nature03412>
- Geng, J., Teferle, F., Meng, X., & Dodson, A. (2010). Kinematic precise point positioning at remote marine platforms. *GPS Solutions*, 14(4), 343–350.
- Han, Y., Zheng, C., & Sun, D. (2016). Signal design for underwater acoustic positioning systems based on orthogonal waveforms. *Ocean Engineering*, 117, 15–21.
- Hayes, S. P., Mangum, L. J., Picaut, J., Sumi, A., & Takeuchi, K. (1991). TOGA-TAO: A moored array for real-time measurements in the tropical Pacific Ocean. *Bulletin of the American Meteorological Society*, 72, 339–347.
- Helber, R. W., Townsend, T. L., Barron, C. N., Dastugue, J. M., & Carnes, M. R. (2013). Validation test report for the Improved Synthetic Ocean Profile (ISOP) system, Part I: Synthetic profile methods and algorithm. Naval Res. Lab. Rep. NRL/MR/7320–13–9364. <https://doi.org/10.21236/ada585251>
- linuma, T., Kido, M., Ohta, Y., Fukuda, T., Tomita, F., & Ueki, I. (2021). GNSS-Acoustic observations of seafloor crustal deformation using a wave glider. *Frontiers in Earth Science*. <https://doi.org/10.3389/feart.2021.600946>
- Ikuta, R., Tadokoro, K., Ando, M., Okuda, T., Sugimoto, S., Takatani, K., et al. (2008). A new GPS-acoustic method for measuring ocean floor crustal deformation application to the Nankai Trough. *Journal of Geophysical Research*. <https://doi.org/10.1029/2006JB004875>
- Japan Coast Guard (2022) GNSS-A data obtained at the SGO-A sites "TOS2", "KUM2", "FUKU", and "MYGI" for the identical four transponder-array. [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.6417480>
- Kido, M., Osada, Y., & Fujimoto, H. (2008). Temporal variation of sound speed in ocean: A comparison between GPS/acoustic and in situ measurements. *Earth, Planets and Space*, 60(3), 229–234. <https://doi.org/10.1186/BF03352785>
- Kido, S., Nonaka, M., & Tanimoto, Y. (2021). Impacts of salinity variation on the mixed-layer processes and sea surface temperature in the Kuroshio-Oyashio confluence region. *Journal of Geophysical Research: Oceans*, 126, e2020JC016914. <https://doi.org/10.1029/2020JC016914>
- LeBlanc, L. R., & Middleton, F. H. (1980). An underwater acoustic sound velocity data model. *Journal of the Acoustical Society of America*, 67, 2055–2062. <https://doi.org/10.1121/1.384448>
- Li, B., Xue, S., Zhao, S., Xiao, Z., Liu, Y., & Yang, C. (2024). GNSS-A geodetic positioning with global gridded marine environment dataset replacing in-field sound speed measurement. *Marine Geodesy*. <https://doi.org/10.1080/01490419.2024.2438113>
- Li, J., Xue, S., Xu, Y., et al. (2023a). Effects of sound speed profile measurement error on underwater positioning. *Journal of Harbin Engineering University*, 44(11), 2062–2070.
- Li, M., Liu, Y., Liu, Y., Chen, G., Tang, Q., Feng, Y., Zhang, L., & Wen, Y. (2023b). Impact of sound travel time modeling on sequential gnss-acoustic seafloor positioning under various survey configurations. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–11.
- Li, Z., He, L., Zhang, R., Li, F., Yu, Y., & Lin, P. (2015). Sound speed profile inversion using a horizontal line array in shallow water. *Science China*

- Physics, Mechanics & Astronomy*, 58, 1–7. <https://doi.org/10.1007/s11433-014-5526-x>
- Liu, Y., Liu, Y., Chen, G., Li, M. (2021). Seafloor single point positioning using GNSS-Acoustic technique with horizontal sound speed gradient estimation. *Scientific Assembly of the International Association of Geodesy*, June 28–July 2, 2021, Beijing, China
- Liu, Y., Li, M., Liu, Y., Chen, G., Tang, Q., Feng, Y., & Wen, Y. (2023). Sequential GNSS-Acoustic seafloor point positioning with modeling of sound speed variation. *Journal of Geodesy*, 97(12), 115. <https://doi.org/10.1007/s00190-023-01810-5>
- Liu, Y., Peng, L., Wu, Y., & Zhou, X. (2006). Calibration of transducer and transponder positions (2006). *Geomatics and Information Science of Wuhan University*, 31(7), 610–612.
- Ma, J., Fang, S., & Zhao, J. (2024). A rigorous real-time acoustic positioning method for ocean bottom seismic exploration. *Satell Navigation* 5(1), 4. <https://doi.org/10.1186/s43020-023-00124-3>
- Mensah, V., & Ohshima, K. I. (2021). Weakened overturning and tide control the properties of Oyashio Intermediate Water, a key water mass in the North Pacific. *Scientific Reports*, 11(1), 14526. <https://doi.org/10.1038/s41598-021-93901-6>
- Metzger, E. J., Smedstad, O. M., Thoppil, P. G., Hurlburt, H. E., Cummings, J. A., Wallcraft, A. J., Zamudio, L., Franklin, D. S., Posey, P. G., Phelps, M. W., Hogan, P. J., Bub, F. L., & DeHaan, C. J. (2014). US Navy operational global ocean and Arctic ice prediction systems. *Oceanography*, 27(3), 32–43. <https://doi.org/10.5670/oceanog.2014.66>
- Ming, F., Yang, Y., & Zeng, A. (2023). Positioning accuracy of GNSS-A in deep sea based on Bayesian estimation. *Chinese Journal of Geophysics*, 66(3), 951–960. <https://doi.org/10.6038/cjg2022P0791>
- Munk, W. H. (1974). Sound channel in an exponentially stratified ocean, with application to SOFAR. *The Journal of the Acoustical Society of America*, 55(2), 220–226. <https://doi.org/10.1121/1.1914492>
- Munk, W., & Wunsch, C. (1979). Ocean acoustic tomography: A scheme for large scale monitoring. Deep Sea Research Part A. *Oceanographic Research Papers*, 26, 123–161. [https://doi.org/10.1016/0198-0149\(79\)90073-6](https://doi.org/10.1016/0198-0149(79)90073-6)
- Naval Research Laboratory (2022) Global Ocean Forecasting System (GOFS) 3.1: 41-layer HYCOM + NCODA Global 1/12° Analysis. <https://www.hycom.org/dataserver/gofs-3pt1/analysis>
- Osada, Y., Fujimoto, H., Miura, S., Sweeney, A., Kanazawa, T., Nakao, S., Sakai, S., Hildebrand, J. A., & Chadwell, C. D. (2003). Estimation and correction for the effect of sound velocity variation on GPS/Acoustic seafloor positioning: An experiment off Hawaii Island. *Earth, Planets and Space*, 10(55), e17–e20. <https://doi.org/10.1186/BF03352464>
- Roemmich, D., Johnson, G. C., Riser, S., Davis, R., Gilson, J., Owens, W. B., et al. (2009). The argo program: Observing the global ocean with profiling floats. *Oceanography*, 22(2), 34–43.
- Sagiya, T., & Thatcher, W. (1999). Coseismic slip resolution along a plate boundary megathrust: The Nankai Trough, southwest Japan. *J. Geophys. Res. Solid Earth*, 104, 1111–1129. <https://doi.org/10.1029/98jb02644>
- Sakic, P., Ballu, V., Crawford, W. C., & Wöppelmann, G. (2018). Acoustic ray tracing comparisons in the context of geodetic precise off-shore positioning experiments. *Marine Geodesy*, 41(4), 315–330.
- Sato, M., Ishikawa, T., Ujihara, N., Yoshida, S., Fujita, M., Mochizuki, M., & Asada, A. (2011). Displacement above the hypocenter of the 2011 Tohoku-Oki earthquake. *Science*, 332(6036), 1395. <https://doi.org/10.1126/science.1207401>
- Spiess, F. (1985). Analysis of a possible sea floor strain measurement system. *Marine Geodesy*, 9(4), 385–398. <https://doi.org/10.1080/15210608509379536>
- Spiess, F. N., Chadwell, C. D., Hildebrand, J. A., Young, L. E., Purcell, G. H., & Dragert, H. (1998). Precise GPS/Acoustic positioning of seafloor reference points for tectonic studies. *Physics of the Earth & Planetary Interiors*, 108(2), 101–112. [https://doi.org/10.1016/S0031-9201\(98\)00089-2](https://doi.org/10.1016/S0031-9201(98)00089-2)
- Watanabe, S. I., Ishikawa, T., Yokota, Y., & Nakamura, Y. (2021). GARPOS: Analysis tool for GNSS-Acoustic seafloor positioning. [Software]. Zenodo. <https://zenodo.org/records/4522027>
- Watanabe, S., Bock, Y., Chadwell, C., Fang, P., & Geng, J. (2017). Long-term stability of the kinematic Precise Point Positioning for the sea surface observation unit compared with the baseline analysis. *Report of Hydrographic and Oceanographic Researches*, 54, 38–73.
- Watanabe, S. I., Ishikawa, T., Yokota, Y., & Nakamura, Y. (2020). GARPOS: Analysis software for the GNSS-A seafloor positioning with simultaneous estimation of sound speed structure. *Frontiers in Earth Science*, 8, Article 597532. <https://doi.org/10.3389/feart.2020.597532>
- Wei, Y., Pei, Y., & Zhang, R. H. (2015). Seasonal variability of the Kuroshio Current at the PN Section in the East China Sea based on in-situ observation from 1987 to 2010. *Acta Oceanologica Sinica*, 34, 12–21. <https://doi.org/10.1007/s13131-015-0662-3>
- Wilson, W. D. (1960). Equation for the speed of sound in sea water. *The Journal of the Acoustical Society of America*, 32(10), 1357–1357. <https://doi.org/10.1121/1.1907913>
- Wu, L., Chen, Z., Lin, X., & Liu, Y. (2020). Building the integrated observational network of “Transparent Ocean.” *Chinese Science Bulletin*, 65(25), 2654–2661. <https://doi.org/10.1360/TB-2020-0558>
- Xu, P., Ando, M., & Tadokoro, K. (2005). Precise, three-dimensional seafloor geodetic deformation measurements using difference techniques. *Earth, Planets and Space*, 57, 795–808. <https://doi.org/10.1186/BF03351859>
- Xue, S., Li, B., Xiao, Z., Sun, Y., & Li, J. (2023). Centimeter-level-precision seafloor geodetic positioning model with self-structured empirical sound speed profile. *Satellite Navigation*, 4(1), 30. <https://doi.org/10.1186/s43020-023-00120-7>
- Xue, S., Yang, Y., Yang, W., & Li, J. (2024). GNSS-A network solution with zenith acoustic delay estimation. *Marine Geodesy*, 47(3), 237–268. <https://doi.org/10.1080/01490419.2024.2307932>
- Yang, F., Lu, X., Li, J., Han, L., & Zheng, Z. (2011). Precise positioning of underwater static objects without sound speed profile. *Marine Geodesy*, 34(2), 138–151. <https://doi.org/10.1080/01490419.2010.518501>
- Yang, H., Guo, X., Miyazawa, Y., Varlamov, S. M., Abe-Ouchi, A., & Chan, W.-L. (2022). Changes in the Kuroshio path, surface velocity and transport during the last 35,000 years. *Geophysical Research Letters*, 49, e2021GL097250.
- Yang, Y., Liu, Y., Sun, D., Xu, T., Xue, S., Han, Y., & Zeng, A. (2020). Seafloor geodetic network establishment and key technologies. *Science China Earth Sciences*. <https://doi.org/10.1007/s11430-019-9602-3>
- Yang, Y., & Qin, X. (2021). Resilient observation models for seafloor geodetic positioning. *Journal of Geodesy*, 95(7), 79. <https://doi.org/10.1007/s00190-021-01531-7>
- Yokota, Y., Ishikawa, T., & Watanabe, S. (2018). Seafloor crustal deformation data along the subduction zones around Japan obtained by GNSS-A observations. *Scientific Data*, 5(1), Article 180182. <https://doi.org/10.1038/sdata.2018.182>
- Yokota, Y., Ishikawa, T., & Watanabe, S. (2019). Gradient field of undersea sound speed structure extracted from the GNSS-A oceanography: GNSS-A as a sensor for detecting sound speed gradient. *Marine Geophysical Research*, 40(4), 493–504.
- Yokota, Y., Ishikawa, T., Watanabe, S., & Nakamura, Y. (2021). Crustal deformation detection capability of the GNSS-A seafloor geodetic observation array (SGO-A), provided by Japan Coast Guard. *Progress in Earth and Planetary Science*, 8(1), 63. <https://doi.org/10.1186/s40645-021-00453-4>
- Yokota, Y., Ishikawa, T., Watanabe, S. I., Tashiro, T., & Asada, A. (2016). Seafloor geodetic constraints on interplate coupling of the Nankai Trough megathrust zone. *Nature*, 534(7607), 374–377. <https://doi.org/10.1038/nature17632>
- Yokota, Y., Watanabe, S. I., Ishikawa, T., & Nakamura, Y. (2022). Temporal change of km-scale underwater sound speed structure and GNSS-A positioning accuracy. *Earth and Space Science*, 9(12), e2022EA002224. <https://doi.org/10.1029/2022EA002224>
- Zhao, J., Liang, W., Ma, J., Liu, M., & Li, Y. (2022). A self-constraint underwater positioning method without the assistance of measured sound velocity profile. *Marine Geodesy*. <https://doi.org/10.1080/01490419.2022.2079778>
- Zhao, S., Wang, Z., He, K., & Ding, N. (2018). Investigation on underwater positioning stochastic model based on acoustic ray incidence angle. *Applied Ocean Research*, 77, 69–77. <https://doi.org/10.1016/j.apor.2018.05.011>

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